

Inflation Inequality Across the Euro Area¹

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October 7, 2026

Abstract

This article documents inflation inequality across household categories in the euro area using harmonized HICP prices and household budget surveys. We construct household and category-level inflation indicators and assess recent fiscal interventions using counterfactual price indices. Household-level inflation dispersion rises sharply during the 2021–2023 episode. Household characteristics explain only 12.4% of within-country variance on average across months, leaving most dispersion within observationally similar household groups. The annual inflation gap between the first and fifth income quintiles peaks at 1.2 percentage points in 2022 and is close to zero in 2021 and 2023. Counterfactual indices indicate that household price policies, including tariff shields, VAT cuts, fuel rebates, and price caps, have effects in 2022 of –1.5 percentage points on annual-average inflation and –0.6 percentage point on the Q1–Q5 inflation gap.

Keywords: Inflation, Distribution of Income, Unconventional Fiscal Policy

JEL codes: E31, D30, C43

¹This paper presents the authors’ views and should not be interpreted as reflecting the views of the Banque de France or the Eurosystem. This paper primarily uses HBS household microdata, supplemented by published tabulations from Eurostat. The responsibility for all conclusions drawn from the data lies entirely with the authors. We thank Pradana Aumars and Lyna El Kamel for excellent research assistance.

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Introduction

This paper documents how the recent inflation surge affected euro-area households. Inflation rates in the euro area rose sharply from mid-2021 onward, displaying considerable dispersion across countries and households. The 2021–2023 episode was a large relative-price shock, initially driven by imported energy and commodity prices.

Inflation inequality emerges when relative-price movements interact with heterogeneous household expenditure shares. If all prices rose proportionally, fixed-basket inflation would be identical across households. We examine how recent energy-price shocks and the fiscal response shaped household inflation exposure across the euro area. Specifically, we construct monthly price indices at the household level and then separately for income, age, and area of residence groups in each euro-area country, following Eurostat’s HICP methodology. We use individual 2020 HBS expenditure records both to construct household-specific indices and to aggregate expenditure into household-group baskets. The household-level indices show dispersion within groups as well as differences between group averages. We also construct counterfactual price indices that remove the main price-based fiscal interventions of 2021–2023. This counterfactual is a direct consumer-price accounting exercise: it isolates the contribution of the measures to household inflation, holding expenditure weights fixed between scenarios and abstracting from behavioral responses.

Our main results are the following. Household-level dispersion rises sharply during the shock. Yet a model saturated with all interactions among income, age, household size, education, and residence explains only 12.4% of within-country variance on average across months, ranging from 8.6% to 17.4%. The remaining 87.6% lies within observationally similar household groups. Thus, income quintiles capture one systematic dimension of heterogeneity rather than the full distribution of household outcomes. The Q1–Q5 inflation gap widens during the shock and varies across euro-area countries. In 2022, the annual Quintile 1–Quintile 5 gap exceeds 3 points in Estonia, Latvia, Lithuania, and the Netherlands and reaches 2.9 points in Ireland. In the euro area, the annual Q1–Q5 inflation gap is 0.1 point in 2021, 1.2 points in 2022, and 0.0 point in 2023 after rounding. Older and rural households experience higher inflation during the rise in energy prices. Household price policies lower annual-average inflation by 0.1, 1.5, and 0.6 percentage points in 2021, 2022, and 2023, respectively. They narrow the annual Q1–Q5 inflation gap by 0.1, 0.6, and 0.5 percentage point in the same years.

A complementary input–output simulation assigns 6.5 percentage points of inflation exposure to the observed extra-EU energy import-price shock and implies a 1.1-point widening of the Q1–Q5 income gap. Household energy makes a contribution of 1.7 points to this gap, partly

offset by vehicle operation and other products. Thus, households are more exposed to the energy shock at the bottom of the income distribution, while the fiscal response primarily cushions the household-energy channel.

The literature measures household inflation by combining price changes with differences in expenditure patterns. For the United States, [Hobijn and Lagakos \(2005\)](#) reweight aggregate price indices for demographic groups. More recently, [Jaravel \(2024\)](#) constructs long-run distributional consumer price indices for the United States, closely following the BLS methodology. Using household-level expenditure or scanner data, [Argente and Lee \(2021\)](#), [Kaplan and Schulhofer-Wohl \(2017\)](#) document household-level dispersion within broad income and demographic groups. Using barcode-level grocery purchases in the United States, [O’Flaherty \(2026\)](#) constructs household-specific price indices that reflect the products households buy and their substitution between them. She shows that demographic group averages mask most household-level inflation dispersion and nearly all of its increase when aggregate inflation rises. For Europe, [Gürer and Weichenrieder \(2018\)](#) combine Household Budget Survey expenditure shares with HICP prices for 25 EU countries and show that household-specific inflation increases measured consumption inequality over 2001–2015. In the euro area, [Ampudia et al. \(2024\)](#) show how household shopping behavior shapes the effect of monetary policy on inflation heterogeneity along the income distribution. [Pallotti et al. \(2024\)](#) combine consumption, income, and balance-sheet data to measure welfare losses across euro-area households. Building on [Charalampakis et al. \(2022\)](#), we take a complementary approach to [O’Flaherty \(2026\)](#). Depending on the exercise, we cover 17 to 20 euro-area countries and the full consumption basket, assigning a common country-product HICP series at the finest harmonized COICOP level supported by each exercise. The household-distribution analysis uses three-digit ECOICOP1 to retain France and Germany in the broad cross-country sample, whereas the policy counterfactuals use four-digit COICOP. This broader scope permits harmonized cross-country comparisons and monthly counterfactuals for price-based fiscal policies. Our RAS-calibrated baskets also aggregate to official HICP item weights. The trade-off is that our measures capture heterogeneity in expenditure shares, but not differences in prices paid, product quality, or substitution within COICOP categories ([Sangani, 2026](#)).

A related literature evaluates the fiscal response to the recent inflation surge. Using household microsimulation models, [Pallotti et al. \(2024\)](#) and [Amores et al. \(2025\)](#) show that fiscal interventions reduced welfare losses and inequality, although broad price measures were less targeted than income support. Using detailed UK household data, [Levell et al. \(2026\)](#) highlight a trade-off between targeting and efficiency: energy price subsidies better protect households with high exposure to energy-price shocks, including within income groups, but distort consumption choices,

while transfers preserve price signals but may be less well targeted. For Germany, [Krebs and Weber \(2024\)](#) examine conditions under which energy price controls can be optimal, while [Dertwinkel-Kalt and Wey \(2025\)](#) show that the design of the energy price brakes can induce suppliers to raise prices while reinforcing households’ incentives to save energy. [Bonnet et al. \(2025\)](#) show, for France, that income-targeted transfers are more effective than fuel price subsidies at limiting welfare losses relative to household income. We instead combine harmonized household inflation measures with monthly counterfactual price paths reconstructed at the COICOP-component level for all 20 euro-area countries. This approach identifies which products drove the distributional effect of price-based policies: lower electricity and gas prices reduce the Quintile 1–Quintile 5 inflation gap, while fuel measures partly offset that reduction because higher-income households spend more on private transport.

The remainder of the paper is organized as follows. Section 1 describes the data and methodology. Section 2 examines developments in inflation inequality in the euro area. Section 3 presents simulations of the heterogeneous effects of unconventional fiscal policies. Section 4 turns to exposure to import-price shocks and the inflation burden.

1 Data and Methodology

Our approach constructs household and household-group price indices using the logic of the Harmonised Index of Consumer Prices (HICP). We first describe the price and expenditure data, sample coverage, product matching, and household groups. We then calibrate expenditure weights to official HICP product weights, construct household and group indices, and aggregate the national results to the euro area.

1.1 Sources, Samples, and Product Coverage

Sources and Samples

The analysis combines public HICP data with restricted Household Budget Survey (HBS) microdata and national HBS inputs. Headline HICP inflation measures price changes for the aggregate household consumption basket. Each national HICP uses price changes for goods and services weighted by household expenditure from national accounts, household budget surveys, and other national sources. Eurostat then aggregates the national indices with country weights that reflect

each country's share of euro-area household consumption expenditure.¹

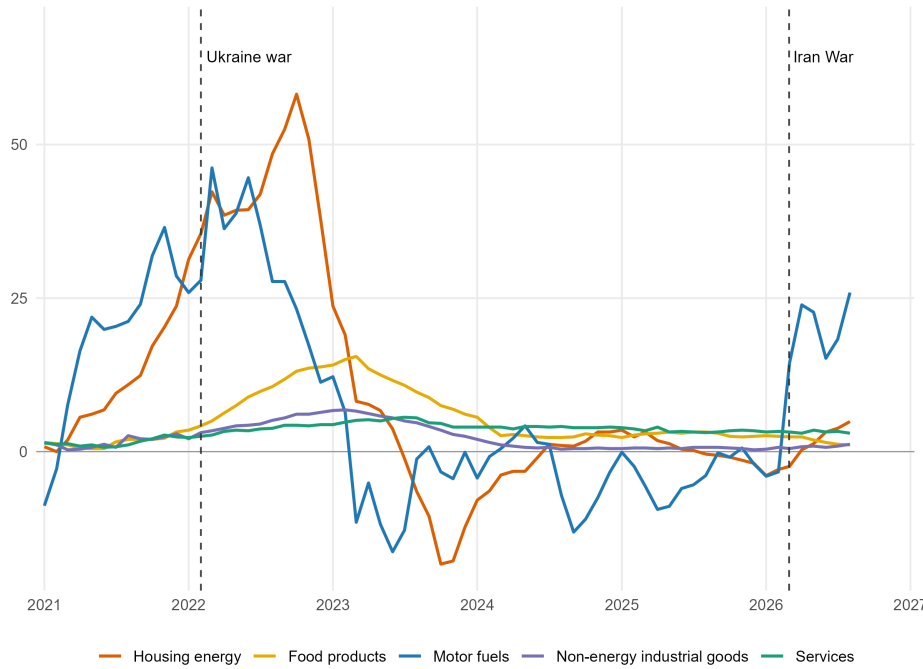
Eurostat updates the HICP product weights annually. The new weights enter with the January release and remain fixed within the calendar year. Let $\mathcal{J}_y$ denote the set of product categories with official HICP weights in year y , $p_{j,y,m}$ the price index for product j in month m , and $p_{j,y,0}$ its price reference for the weight year, corresponding to the previous December. The official within-year Laspeyres link and chained index are

$$P_{y,m} = \sum_{j \in \mathcal{J}_y} \omega_{j,y}^{\text{HICP}} \frac{p_{j,y,m}}{p_{j,y,0}}, \quad (1)$$

$$I_{y,m} = I_{y-1,12} P_{y,m}. \quad (2)$$

Figure 1 shows inflation across the main HICP components in the euro area since January 2021. Energy and fuel prices rise sharply from mid-2021 and peak in 2022. Inflation subsequently declines across most components, with energy prices falling more sharply than food prices and other components.

Figure 1: Inflation developments in the euro area (year on year, in %)



Sources: Eurostat HICP euro-area aggregate; authors' calculations.

¹The calculation follows Regulation (EU) 2016/792 and common data-collection practices. HICP products follow the European Classification of Individual Consumption according to Purpose (ECOICOP), and historical series are mapped to the classification used in the HBS. National statistical institutes compile the HICP in collaboration with Eurostat.

Our primary expenditure source is the 2020 HBS microdata. We use household expenditure records for 19 euro-area countries, excluding Italy. The records contain expenditure by product, total net household income, and household characteristics. Group-level indicators cover all 20 euro-area countries through the national and tabulation-based exceptions described below. Household distributions cover 152,613 households in 18 countries: Italy is excluded because its retained national input supplies group baskets rather than household records, and Portugal because its individual expenditure divisions cannot separate rents. These households remain fixed across price years, so the monthly series does not represent a panel of newly observed consumption choices. Figure 3 follows this sample from January 2021 to December 2025.

The 2020 HBS wave combines surveys collected over country-specific reference periods. Field-work dates and survey frequency therefore need not coincide across countries. The fixed wave provides a cross-section of consumption heterogeneity rather than annual observations on changing household baskets.² Repeated Spanish HBS waves suggest that broad differences in expenditure shares between low- and high-income households remain relatively stable even when aggregate consumption patterns change (Appendix C).

Product Coverage and Treatment of Rents

The empirical exercises use the finest harmonized product detail supported by their source data. The household-distribution analysis underlying Figures 3 and 4 matches HICP prices and HBS expenditure at the three-digit ECOICOP1 level. This is the finest common household-level detail that retains France and Germany in the broad cross-country sample; countries with finer HBS expenditure codes are aggregated to this level. Source price series must be complete over the full calculation period before this aggregation, which keeps the product composition within each three-digit category fixed over time. Group-level exercises and policy counterfactuals retain four-digit detail where their inputs support it, using the nearest common parent-category price when a four-digit HICP series is unavailable. Historical and validation exercises retain their explicitly stated matching levels. Appendix A compares the measured Q1–Q5 inflation gap across two- to five-digit COICOP levels and reports the matching level used in each specification.

The baseline excludes actual and imputed rents and removes the corresponding HICP components before matching. Parent categories are then rebuilt, and the remaining HBS expenditure and HICP product weights are normalized over the common non-rent support. This coverage applies to household indices, group indices, and policy counterfactuals. The resulting measures capture

²See [Accardo et al. \(2023\)](#) on estimating changes in consumption inequality between survey waves.

non-rent consumption-price exposure rather than an all-items cost-of-living index. Table F.1 reports alternative rent conventions, while Appendices F and H report the corresponding normalization and validation exercises.

Household Groups and National Exceptions

We construct groups by income, age of the household reference person, and area of residence. The HBS also records education, which enters the household-level variance decomposition. Residence groups use the Degree of Urbanisation (DEGURBA).³

We use HBS survey weights throughout the construction of nationally representative group baskets. Income groups rank households by total net household income without adjustment for household size. They therefore describe household income rather than living standards per consumption unit. Evidence from the United States shows why this distinction matters. Using Consumer Expenditure Survey data, [Klick and Stockburger \(2024\)](#) find that equivalising income substantially changes the demographic composition of income quintiles: the lowest quintile contains fewer one-person households and more households with children. Yet the resulting changes in income-group inflation rates are small over their 2006–2023 comparison period. Equivalisation therefore makes households within income groups more comparable in terms of living standards without necessarily changing the qualitative inflation gradient. National income-quintile thresholds are defined separately within each national HBS sample and survey wave. Missing income excludes a household from income baskets but not from age, residence, or unconditional analyses when the required variables are available. Appendix G compares this definition with equivalised income while holding household inflation rates, the sample, and weights fixed.

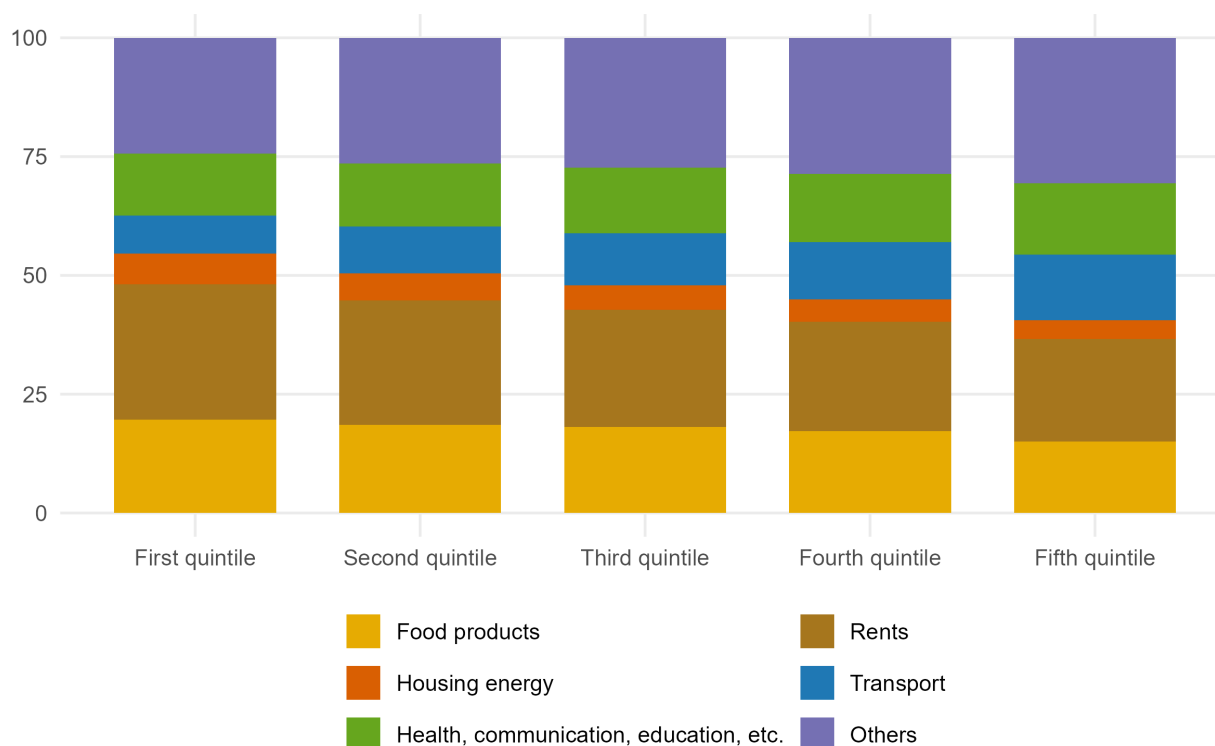
The main 2021–2023 analysis uses fixed 2020 HBS baskets. For each national group, we aggregate individual product expenditure into a representative consumption basket. Three national exceptions preserve broader group coverage. Italy uses the national group baskets described in Appendix E. Portugal uses published group tabulations for non-rent inflation because its individual expenditure divisions cannot isolate rents. Malta uses published residence-area baskets and relative-expenditure weighting because the required groups and RAS margins are unavailable.

Figure 2 compares income-quintile baskets at the euro area level. For descriptive completeness,

³DEGURBA uses a common European methodology based on a 1 km² population grid. It distinguishes cities (densely populated areas), towns and suburbs (intermediate-density areas), and rural areas (thinly populated areas). Urban centres have at least 50,000 inhabitants and 1,500 inhabitants per km²; urban clusters have at least 5,000 inhabitants and 300 inhabitants per km²; the remaining areas are rural.

the figure includes actual and imputed rents, although both are excluded from the inflation indices because their treatment is not sufficiently harmonized across countries. Lower-income households allocate larger expenditure shares to food and housing. Within housing, household energy is retained in the inflation indices and is central to the analysis. Exposure to a given price shock therefore depends on both these basket differences and the products affected by the shock. Housing energy and private-transport expenditure are also higher in rural areas than in cities, reflecting differences in heating needs and mobility constraints.

Figure 2: Consumption baskets by income quintile in the euro area (in % of total expenditure)



Notes: Country budget shares are averaged using 2020 HICP country weights, normalized across the 20 countries and held common across quintiles. These are national income quintiles, not euro-area income ranks. Rents comprise actual and imputed rents; the inflation indices exclude both. Microdata supply the baskets except for the documented national and tabulation-based fallbacks.

Sources: HBS microdata, Eurostat, Istat, authors' calculations.

1.2 Calibration and Price Indices

RAS Calibration

Unadjusted HBS expenditure shares do not generally aggregate to the annual HICP product weights. We therefore use HBS expenditure to distribute each official product total across households and groups. Row and Column Adjustment Scaling (RAS) preserves the relative HBS expenditure profiles while imposing margins consistent with the HICP construction.

For household i in country c , let a_{ic} denote its survey weight and $E_{ic,y}$ its retained expenditure on the matched non-rent product set $\mathcal{J}_{c,y}$. Let $\omega_{c,j,y}^{\text{HICP}}$ be the official weight of product j in year y . We normalize the household expenditure and HICP product margins to sum to one:

$$s_{ic,y} = \frac{a_{ic} E_{ic,y}}{\sum_h a_{hc} E_{hc,y}}, \quad \bar{\omega}_{c,j,y}^{\text{HICP}} = \frac{\omega_{c,j,y}^{\text{HICP}}}{\sum_{k \in \mathcal{J}_{c,y}} \omega_{c,k,y}^{\text{HICP}}},$$

where h indexes households in country c . RAS rescales the household–product expenditure matrix to match both margins. Its calibrated cells $M_{j,ic,y}$ satisfy

$$\sum_{j \in \mathcal{J}_{c,y}} M_{j,ic,y} = s_{ic,y}, \quad \sum_i M_{j,ic,y} = \bar{\omega}_{c,j,y}^{\text{HICP}}.$$

Normalizing each household’s calibrated expenditure gives its product weights:

$$\omega_{j,ic,y} = \frac{M_{j,ic,y}}{s_{ic,y}}, \quad \sum_{j \in \mathcal{J}_{c,y}} \omega_{j,ic,y} = 1.$$

The expenditure-weighted average of household baskets therefore reproduces the normalized HICP product shares on the matched support.

For a household group g , the calibration replaces the individual margin with the group’s share $s_{g,y}$ of total HBS expenditure. If $M_{j,g,y}$ is the calibrated product–group mass, the group basket is $\omega_{j,g,y} = M_{j,g,y}/s_{g,y}$. It sums to one, and its expenditure-weighted average across groups reproduces the normalized HICP product shares. Appendix F details the RAS algorithm and the matching of group margins to HBS waves.

Household-Level Price Indices

Let $p_{c,j,y,m}$ denote the product price index in month m of year y . Product indices are first expressed relative to the preceding December, aggregated with annual household weights, and then chained

to the preceding December's household index:⁴

$$P_{ic,y,m} = \sum_{j \in \mathcal{J}_{c,y}} \omega_{j,ic,y} \frac{p_{c,j,y,m}}{p_{c,j,y-1,12}}, \quad (3)$$

$$I_{ic,y,m} = I_{ic,y-1,12} P_{ic,y,m}. \quad (4)$$

The household application uses fixed 2020 expenditure profiles, while annual HICP margins update aggregate product composition.

Group-Level Price Indices

For each income, age, or residence group, we aggregate the HBS expenditure records before applying RAS. Group indices therefore need not equal averages of separately calibrated household indices. Suppressing country subscripts, the within-year link and chained group index are

$$P_{g,y,m} = \sum_{j \in \mathcal{J}_y} \omega_{j,g,y} \frac{p_{j,y,m}}{p_{j,y-1,12}}, \quad I_{g,y,m} = I_{g,y-1,12} P_{g,y,m}. \quad (5)$$

Here $\mathcal{J}_y$ is the matched non-rent product set. The baseline uses the 2020 HBS wave throughout; historical exercises using successive waves retain their stated matching convention.

1.3 National and Euro-Area Aggregation

We use two aggregation procedures for two distinct objects. First, the distributional bands in Figure 3 summarize national household-inflation distributions. Within each country, we calculate the mean and the 10th, 20th, 80th, and 90th empirical percentiles of household year-on-year inflation. We then average each national statistic using Eurostat's annual HICP country expenditure weights. For a national statistic $s_{c,y,m}$, the reported series is

$$s_{y,m} = \sum_c \omega_{c,y} s_{c,y,m}, \quad \omega_{c,y} = \frac{W_{c,y}}{\sum_{k \in \mathcal{C}} W_{k,y}},$$

where $W_{c,y}$ is the published all-items HICP country weight and $\mathcal{C}$ is the fixed country sample. These bands average national percentile endpoints; they are not percentiles of a pooled euro-area distribution. The country weights are not additionally adjusted for rents.

Second, aggregate household and group price indices require aggregation of December-relative price links before chaining. Let $q_{c,y}$ denote the annual HICP country expenditure weight after

⁴See Eurostat's `hicp` R package, which implements the Eurostat HICP aggregation.

removing rent expenditure and normalizing over the included countries, so $\sum_c q_{c,y} = 1$. The household aggregate is

$$P_{c,y,m} = \frac{\sum_i a_{ic} E_{ic,y} P_{ic,y,m}}{\sum_i a_{ic} E_{ic,y}}, \quad (6)$$

$$P_{y,m} = \sum_c q_{c,y} P_{c,y,m}, \quad I_{y,m} = I_{y-1,12} P_{y,m}. \quad (7)$$

The same sequence applies to group indices using group-specific expenditure masses. Averaging household inflation rates directly does not implement this annual chain aggregation. All baseline inflation results exclude actual and imputed rents. Official headline series and the explicitly identified rent-sensitivity and validation exercises retain their stated comparison coverage.

2 Main Results

This section presents our main results at the euro-area level in the wake of the 2021–2023 energy crisis. We further document cross-country and cross-product heterogeneity in inflation inequality.

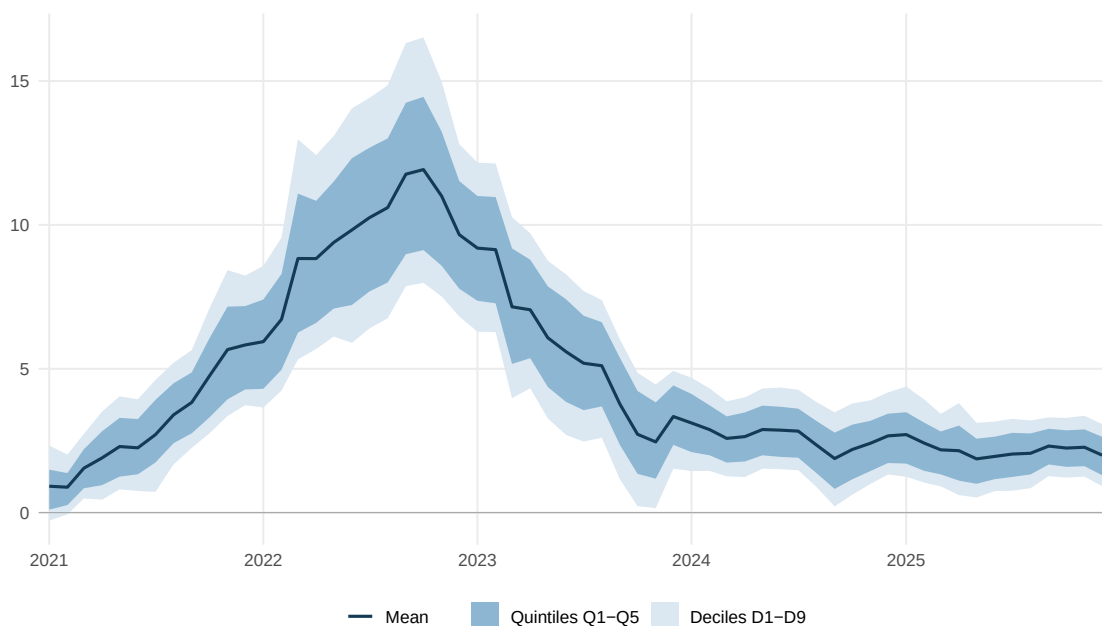
2.1 Inflation Inequality in the Euro Area

2.1.1 Household-Level Inflation Dispersion

Figure 3 shows country-weighted national means and quintile and decile ranges from January 2021 to December 2025. The mean rises from 0.9% in January 2021 to a peak of 11.9% in October 2022. In that month, the weighted national D1 and D9 cutoffs are 8.0% and 16.5%, respectively. The inflation surge therefore coincides with greater dispersion across household baskets within countries.

The band widths summarize within-country basket differences, averaged using annual country expenditure weights. Although the underlying household sample is fixed, the households located at each national cutoff can change from month to month. The bands therefore describe the cross-sectional distribution in each month rather than the inflation path of a fixed set of households at those cutoffs. Moreover, unlike the transaction data used by [Kaplan and Schulhofer-Wohl \(2017\)](#), our data do not capture differences in prices paid by households for the same product within a country.

Figure 3: Household inflation: country-weighted national mean, quintile and decile ranges, 2021–2025 (year on year, in %)



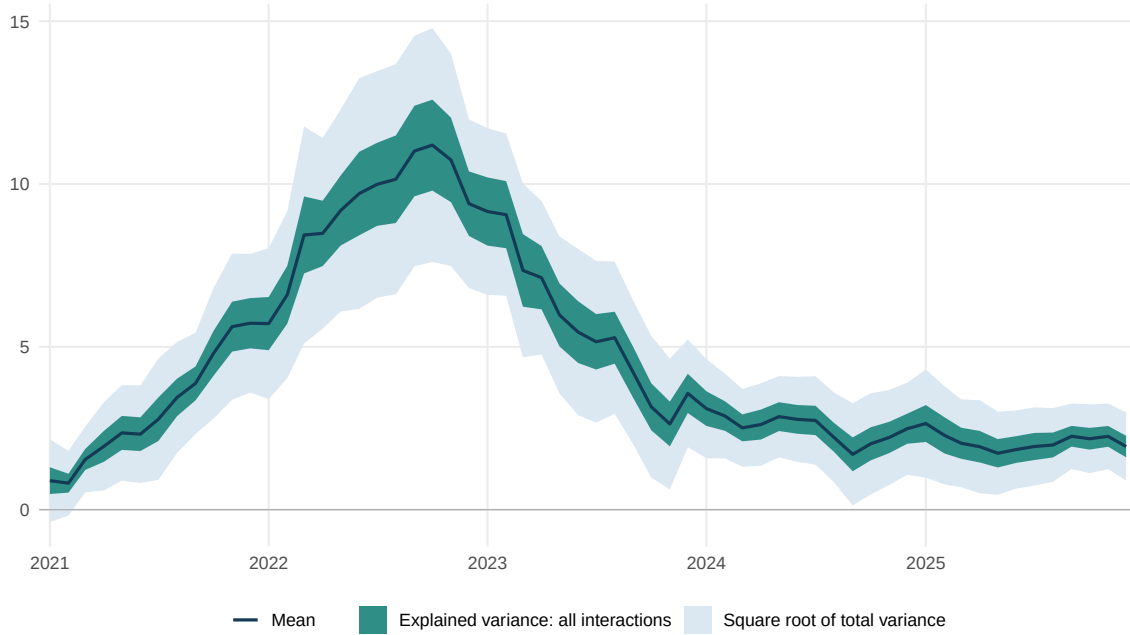
Notes: Monthly year-on-year inflation for 152,613 households in 18 euro-area countries, using fixed 2020 HBS microdata baskets. Italy and Portugal are excluded because household baskets or the detail needed to remove rents are unavailable. Actual and imputed rents (041 and 042) are excluded. Household inflation is constructed at the three-digit ECOICOP1 level. Household indices use annual RAS calibration with expenditure margins at that level. National means and distributional cutoffs first use HBS survey weights, without an income filter. Each national statistic is then averaged using annual Eurostat HICP country weights normalized over these 18 countries. The solid line shows the mean; the darker and lighter bands show averages of national Q1–Q5 and D1–D9 ranges. These are neither pooled euro-area ranges nor confidence intervals.

Sources: Eurostat HICP, HBS 2020 microdata, authors' calculations.

Figure 4 shows the square roots of total and explained household inflation variance. For each country and month, we estimate a saturated model of national income quintile, age and education of the household reference person, household size, and degree of urbanization.

We average national R^2 values using annual Eurostat HICP country weights. The explained share averages 12.4% and ranges from 8.6% to 17.4% across months. The outer band extends one total standard deviation, $\sqrt{V}$, above and below mean inflation, while the darker band has half-width $\sqrt{R^2 V}$. Both bands widen during the 2021–2023 inflation surge, but even this flexible model leaves 87.6% of variance within observationally similar groups.

Figure 4: Household inflation: explained and total variance with saturated household-characteristic interactions, 2021–2025



Notes: Separate unweighted saturated models are estimated for each country and month using 138,798 households in 17 countries. Household inflation is constructed at the three-digit ECOICOP1 level. Relative to the 18-country sample in Figure 3, the Netherlands is excluded because education is unavailable. Cells are defined by the full interaction of national income quintile, age and education of the household reference person, household size, and degree of urbanization. No product variables enter the model. National R^2 values and within-country variances are aggregated using annual HICP country weights normalized over the sample. The solid line is mean inflation. The darker band has half-width $\sqrt{R^2 V}$, and the full outer band has half-width $\sqrt{V}$. The bands are descriptive, not confidence intervals.

Sources: Eurostat HICP, HBS 2020 microdata, authors' calculations.

Household inflation dispersion moves closely with mean inflation. Following O'Flaherty (2026), we ask whether this co-movement comes from widening inflation differences between observable demographic cells or from greater dispersion within those cells. Define the share of the correlation accounted for by between-cell differences as

$$S = 100 \left[1 - \frac{\text{Corr}(\bar{\pi}_t, V_t^W)}{\text{Corr}(\bar{\pi}_t, V_t)} \right],$$

where V_t is total household inflation variance and V_t^W is the residual variance within demographic cells. From 2021 to 2025, the correlation between mean inflation and total variance is 0.945. When the between-cell component is removed and only within-cell variance is retained, the correlation is 0.940. Thus, 99.5% of the original correlation remains ($0.940/0.945 = 0.995$), and between-cell

differences account for only $S = 0.5\%$ under this diagnostic. Over 2021–2023, 99.2% remains (0.914/0.922), giving $S = 0.8\%$. This result is distinct from the variance decomposition above: 87.6% is the average share of variance located within demographic cells, whereas 99.5% is the share of the time-series association between mean inflation and dispersion that remains after between-cell differences are removed. Together, the two results show that observed demographic characteristics explain little of the level of household inflation dispersion and almost none of its rise and fall over time.

The peak-month gaps convey the same distinction. In October 2022, the Q1–Q5 range of the household-level inflation distribution reached 5.3 percentage points, more than twice the 2.0-point gap between the mean inflation rates of the first and fifth income quintiles. Income-group averages therefore identify a systematic exposure gradient, but capture only a limited part of the dispersion across households.

Table 1 compares separate one-way classifications with the fully interacted demographic cells. Income has the highest stand-alone explanatory power, accounting for 2.5% of household inflation variance on average, while no other single characteristic accounts for more than 1.8%. These estimates are not additive: each column defines a separate partition of households, whereas the last column also captures the explanatory power of combinations of characteristics. None of the individual classifications accounts for more than 0.4% of the association between mean inflation and cross-sectional variance, and their full interaction accounts for only 0.5%.

The 12.4% share is close to the 11.6% reported by O’Flaherty (2026) for US grocery inflation. The two estimates are conceptually aligned because both compare total variance with the variance remaining within fully crossed demographic cells. O’Flaherty’s calculation is a cell-mean decomposition, mathematically equivalent to a saturated regression with categorical indicators, rather than a structural or causal model. The levels are not directly comparable, however, because the studies differ in their demographic variables, cell definitions, periods, consumption coverage, product granularity, price measurement, and weighting. In O’Flaherty, the within-cell component also captures barcode-level differences in products purchased and substitution. In our setting, all households in a country face common COICOP price changes, so the within-cell component reflects differences in expenditure shares across product categories. It is therefore economically meaningful exposure heterogeneity not summarized by the observed characteristics, rather than measurement error or economically unexplained variation.

Table 1: Household inflation variance and its association with mean inflation explained by observed characteristics

	Income	Age	Education	Household size	Urbanization	All interactions
Mean share of variance explained (%)	2.5	1.8	1.5	1.5	1.6	12.4
Share of inflation–dispersion correlation explained (%)	0.4	0.2	0.1	0.1	-0.2	0.5

Notes: Separate unweighted group-mean models are estimated for each country and month from January 2021 to December 2025 using the common complete-case sample of 138,798 households in 17 countries. Household inflation is constructed at the three-digit ECOICOP1 level. Each of the first five columns classifies households only by the indicated characteristic; these columns are separate one-way partitions and are not additive. The last column uses the full interaction of income quintile, age, education, household size, and degree of urbanization. The first row averages the country-weighted national shares of variance explained across months. The second row reports the percentage reduction in the correlation between mean inflation and total variance when total variance is replaced by within-group variance. A negative value means that removing between-group differences slightly increases this correlation. Country weights are annual HICP weights normalized over the sample. The exercise is descriptive and does not assign a causal interpretation to household characteristics.

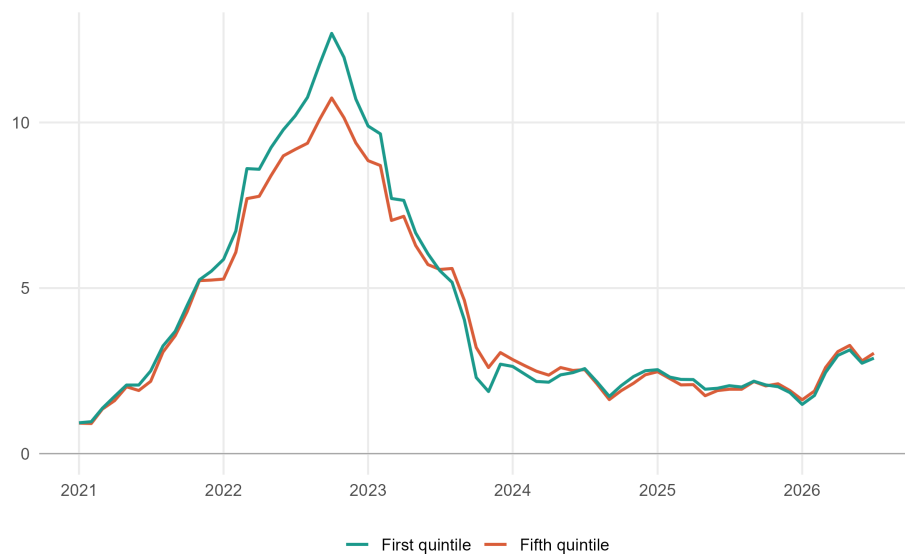
2.1.2 Inflation Inequality by Household Groups

Although observable characteristics jointly explain only a small share of household-level inflation dispersion, group averages reveal systematic differences in exposure during the inflation surge. We consider three dimensions, income, the age of the household reference person, and area of residence, to document which broad household groups faced higher inflation. These comparisons describe average exposure gradients rather than the full distribution of household outcomes.

The inflation gap between households in the bottom and top 20 percent of their national income distributions reached 1.2 percentage points in 2022, its highest annual value in our series since 2000.⁵ Figure 5 traces the monthly dynamics behind this annual result. From January 2011 through September 2021, the gap remained between -0.25 and 0.25 percentage point in about four out of five months, consistent with the low-inflation environment. It then rose from 0.1 percentage point in September 2021 to a monthly peak of 2.0 percentage points in October 2022 before narrowing as energy inflation receded. This time profile is consistent with recent evidence from [Bobasu et al. \(2023\)](#), although differences in weighting and country coverage may affect the measured size of the gap.

⁵Appendix I presents the corresponding long-run price indices.

Figure 5: Inflation inequality in the euro area, by income quintile (year on year, in %)



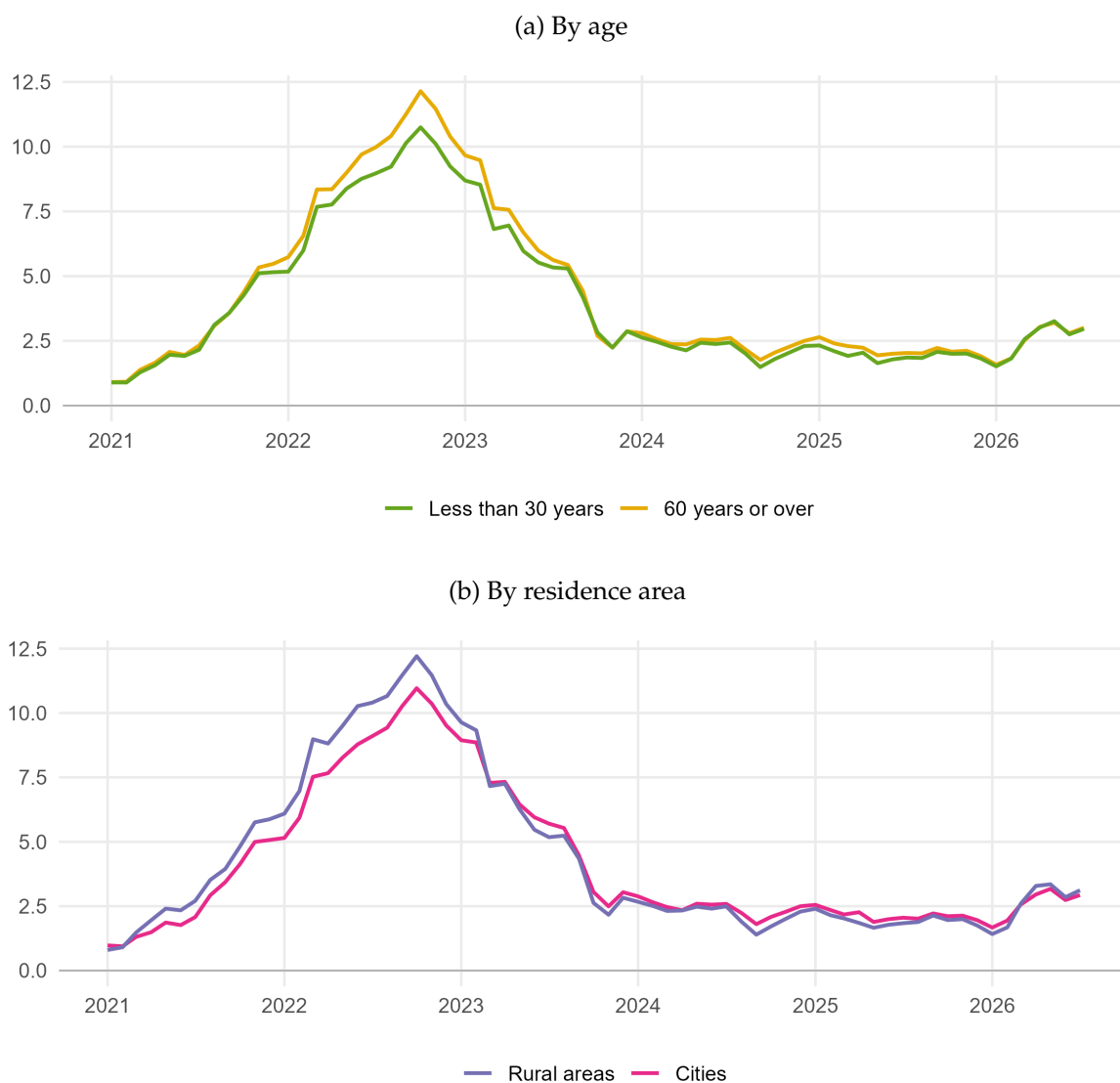
Notes: This figure reports year-on-year inflation rates for the first and fifth income quintiles in the euro area from January 2021 to July 2026. Group baskets are constructed from fixed 2020 HBS microdata, using total net household income without equivalisation, with the Italian approximation and Portuguese tabulation fallback described in Section 1. The country series are aggregated using annual HICP country weights.

Sources: Eurostat HICP, HBS, authors' calculations.

Income is not the only systematic dimension of inflation exposure. Figure 6 compares inflation rates by age and residence area. The age gap is largest near the inflation peak in late 2022: households whose reference person is aged 60 or older face inflation about 1.3 percentage points higher than households whose reference person is younger than 30. The two series converge as energy inflation recedes, showing that the age gradient is concentrated in the energy-shock period rather than persistent throughout the sample.

The residence-area gradient follows a similar time profile and reflects differences in the structure of energy demand. Rural households tend to spend a larger share of their budget on transport fuels and home energy, partly because commuting distances are longer and the set of available transport substitutes is more limited. This raises their exposure when energy prices accelerate, but also makes their measured inflation fall faster when those price pressures unwind. [Gautier and Montornes \(2024\)](#) obtain similar results for the four largest euro-area countries under slightly different assumptions.

Figure 6: Inflation by household categories in the euro area (year on year, in %)



Notes: Year-on-year inflation from fixed 2020 HBS group baskets and annual HICP country weights, excluding rents. National exceptions are documented in Section 1.
Sources: Eurostat HICP and HBS, authors' calculations.

2.2 Product and Country Heterogeneity During the 2022–2023 Energy Shock

EA20 averages also show heterogeneity across products and countries. The Q1–Q5 inflation gap can be interpreted as a weighted covariance between relative differences in expenditure shares and product-level inflation. A product contributes positively to the gap when it is relatively more important in the Q1 basket and its price increases. Conversely, a negative contribution means that the product raises Q5 inflation more than Q1 inflation; it does not necessarily indicate a decline in its price. Table 2 reports the product-level terms of this decomposition. In 2022, household energy accounts for 3.1 points of average inflation and contributes 1.9 points to the 1.2-point gap; food adds 1.9 points to average inflation and 0.7 point to the gap. These positive distributional contributions are partly offset by personal transport, other transport, and other products, which contribute -0.4 , -0.3 , and -0.5 percentage points to the gap. Higher-income households devote larger expenditure shares to personal transport, other transport, and the broad “other” category. Because prices in other transport and other products rise more slowly than those of energy and food, however, their negative contributions offset only part of the energy- and food-driven Q1 disadvantage. In 2023, the positive food contribution is again offset by other product groups, leaving a zero aggregate gap after rounding.

Table 2: Product-group contributions to annual-average euro-area inflation and Q1–Q5 inflation inequality, 2021–2023

Product group	Weight (2022, %)	Average inflation			Inflation inequality		
		2021	2022	2023	2021	2022	2023
Food and non-alcoholic beverages	17.8	0.3	1.9	2.1	0.1	0.7	0.7
Alcoholic beverages, tobacco and narcotics	4.6	0.1	0.2	0.3	0.1	0.1	0.2
Clothing and footwear	5.6	0.1	0.1	0.2	0.0	0.0	-0.1
Housing maintenance and furnishings	8.0	0.2	0.6	0.5	-0.1	-0.3	-0.2
Water, electricity, gas and other fuels	6.8	0.7	3.1	0.2	0.4	1.9	0.1
Operation of personal transport equipment	6.3	0.8	1.4	0.1	-0.3	-0.4	0.0
Transport (excluding vehicle operation)	9.6	0.1	0.4	0.3	-0.1	-0.3	-0.2
Other	41.1	0.4	1.3	1.9	-0.1	-0.5	-0.6
Total	100.0	2.8	9.0	5.6	0.1	1.2	0.0

Notes. Product rows contribute to annual EA20 average inflation (percent) and the Q1–Q5 gap (percentage points).

Annual rates compare annual-average indices with the preceding year. Weights are 2022 EA20 expenditure shares.

Sources: Eurostat, authors' calculations.

Cross-country differences reflect variation in energy mixes, household consumption patterns, and fiscal responses. These factors shaped how the energy-price shock, which began in 2021 and intensified, especially in natural gas markets, after Russia's invasion of Ukraine, translated into household inflation.

Table 3 documents this cross-country heterogeneity by reporting annual-average inflation and the annual Quintile 1–Quintile 5 inflation gap. The gap is largest in 2022 in most countries: it exceeds 3 points in Estonia, Latvia, Lithuania, and the Netherlands and reaches 2.9 points in Ireland. Several countries subsequently record a negative gap in 2023 as energy-price pressures recede, while the euro-area gap falls from 1.2 points in 2022 to zero after rounding in 2023. This cross-country pattern is qualitatively similar to [Claeys et al. \(2022\)](#), who also find larger inflation gaps between low- and high-income households in countries with higher inflation.

The same interaction between price dynamics and expenditure shares operates across countries. A price increase in a product category generates inflation inequality only to the extent that the affected category has different expenditure weights across household groups. Large aggregate

shocks may generate limited inequality when price increases fall on categories with similar weights across quintiles.

Table 3: Annual-average inflation rates and Q1–Q5 inflation inequality in the 20 euro-area countries, 2021–2023

Country	Weight (2022, %)	Average inflation			Inflation inequality		
		2021	2022	2023	2021	2022	2023
Germany	28.1	3.5	9.7	6.5	-0.4	1.4	0.5
France	20.4	2.3	6.6	6.1	0.4	0.2	0.2
Italy	16.3	2.0	9.0	6.0	0.1	1.6	0.5
Spain	11.0	3.1	8.6	3.4	0.8	0.9	-0.7
Netherlands	5.4	3.2	13.2	4.0	0.0	3.1	-1.2
Belgium	4.0	3.4	11.0	2.0	1.3	2.3	-2.7
Austria	3.3	2.8	9.1	7.7	-0.5	0.0	0.5
Portugal	2.2	0.9	8.3	5.3	0.8	0.2	-0.9
Greece	2.2	0.6	9.6	4.1	0.3	1.2	-0.6
Finland	1.9	2.2	7.8	4.5	-0.2	-0.9	0.4
Ireland	1.5	2.4	7.9	5.0	0.5	2.9	1.1
Slovakia	0.8	2.9	12.3	11.1	-0.9	0.1	0.5
Croatia	0.7	2.7	10.8	8.5	0.0	0.2	-0.3
Lithuania	0.6	4.7	18.9	8.7	0.2	3.3	0.4
Slovenia	0.4	2.1	9.4	7.2	0.0	0.4	1.5
Luxembourg	0.3	3.6	8.6	3.0	0.1	0.2	-0.1
Latvia	0.3	3.3	17.6	9.1	-0.2	4.5	2.5
Estonia	0.2	4.5	19.4	9.4	0.6	4.8	1.3
Cyprus	0.2	2.3	8.2	3.9	0.0	0.8	0.2
Malta	0.1	0.8	6.1	5.4	-0.2	0.3	0.6
Euro area	100.0	2.8	9.0	5.6	0.1	1.2	0.0

Notes. Annual inflation compares annual-average indices with the preceding year. The inflation gap is Q1 minus Q5, in percentage points. Weights are 2022 EA20 HICP shares. Sources: Eurostat, authors' calculations.

These observed gaps combine differences in household baskets, national price dynamics, and policy interventions. Having documented the first two channels, the next section isolates the direct contribution of household price policies.

3 Contributions of Unconventional fiscal Policies to Inflation Inequality

This section focuses on policies implemented during the 2021–2023 energy crisis that directly changed consumer prices. They include electricity and gas price caps and shields, VAT and energy-tax reductions, and fuel rebates.

A conventional fiscal response would tighten the budget to cool aggregate demand and inflation, potentially reducing transfers to households. Unconventional fiscal measures instead alter retail prices directly and can temporarily lower measured inflation while supporting household purchasing power (Dao et al., 2023).

3.1 Constructing Counterfactual Price Indices

In response to the energy crisis, euro-area governments introduced a broad array of measures to mitigate household exposure to rising energy costs (Arregui et al., 2022, Dao et al., 2023, García-Miralles, 2023). We compare observed, policy-inclusive retail prices with counterfactual price paths that remove the covered energy and food price measures. This comparison isolates the measures’ effects on household-specific inflation by holding consumption quantities fixed and allowing only prices to differ between the “with-policy” and “without-policy” scenarios. It is therefore an accounting counterfactual rather than a behavioral one: it does not capture quantity adjustments, such as the observed decline in residential energy consumption during the crisis. Income transfers and firm-only measures are excluded because they do not directly alter the covered consumer prices.

We combine the Bruegel energy-measures dataset⁶ with national sources to classify price measures, adding Spanish and Portuguese food-product VAT cuts. Table D.1 in the appendix reports the inventory.

The available information determines how we construct each price path. First, direct reconstructions use published reference tariffs, statutory tax rates, or statutory unit rebates, with coverage and pass-through specified. Second, calibrations choose a measure-specific intensity parameter so that the geometric mean of monthly counterfactual-to-observed price ratios over the active policy months matches the assumed component-level ratio in the measure-level assumptions file. These targets summarize coverage and pass-through using the policy inventory and its cited national sources; they are assumptions rather than observed counterfactual prices. Third, conditional tariff or bill benchmarks apply statutory rules to a reference contract or representative bill when

⁶<https://www.bruegel.org/dataset/national-policies-shield-consumers-rising-energy-prices>

household contract shares or consumption quantities are unavailable. Their results depend on the chosen reference contract and consumption quantity. They also depend on assumed policy exposure. These three procedures classify the information used to construct each counterfactual price path; the subsections below instead organize measures by policy instrument. Each instrument subsection combines the general price adjustment with representative country implementations and their limitations.

Counterfactual price indices are constructed at the four-digit COICOP level by replacing each observed HICP component index with its counterfactual counterpart. Let $p_{j,t}$ denote the observed HICP component index for product j in month t , and let $\tilde{p}_{j,t}$ denote the corresponding counterfactual index. Here t indexes calendar months, corresponding to the year-month pair (y, m) used above. Let $r_{j,t}$ denote the observed retail price per unit and $\tilde{r}_{j,t}$ its counterfactual counterpart, expressed in the same monetary units. For each simulated measure, we apply the implied price ratio to the observed component index⁷:

$$\tilde{p}_{j,t} = p_{j,t} \frac{\tilde{r}_{j,t}}{r_{j,t}}. \quad (8)$$

VAT and Non-Fuel Excise-Tax Reductions. For VAT reductions, the observed retail price includes the temporary tax rate. We recover the counterfactual price by removing that rate and applying the standard rate:

$$\tilde{r}_{j,t} = r_{j,t} \frac{1 + \tau_{j,t}^0}{1 + \tau_{j,t}^1}, \quad (9)$$

where $\tau_{j,t}^0$ is the standard VAT rate and $\tau_{j,t}^1$ is the temporary rate. Non-fuel ad valorem excises follow the same ratio logic. Road-fuel excise reductions are classified with other fuel-price reductions and use the unit-price adjustment described below. Coverage and pass-through are calibrated where the statutory information is incomplete. France's electricity-tax cuts and Spain's energy-tax reductions are reconstructed from the corresponding statutory rates. Spanish fuel rebates are treated as unit rebates, as described below.

Administered Prices. For regulated-price shields and caps, we use an external reference path where available, such as a commodity-price benchmark or a tariff published by the national regulator (for example, the CRE in France):

$$\tilde{r}_{j,t} = r_{j,t}^{\text{ref}}. \quad (10)$$

⁷When several measures apply to the same product in the same country and month, the R scripts combine their counterfactual-to-observed price ratios by adding their deviations from one to a baseline ratio of one.

When the observed tariff and the counterfactual reference tariff are both available, the counterfactual retail price is obtained by rescaling the observed price. If $T_{j,t}^{\text{obs}}$ denotes the regulated tariff actually applied and $T_{j,t}^{\text{ref}}$ the tariff that would have applied without the intervention, then:

$$r_{j,t}^{\text{ref}} = r_{j,t} \frac{T_{j,t}^{\text{ref}}}{T_{j,t}^{\text{obs}}}. \quad (11)$$

For France, published regulated tariffs provide the reference paths for the gas and electricity price shields. For Spain, regulated gas-price limits and the Iberian mechanism use documented reference paths where available and calibrated paths otherwise. For the Netherlands, we use the broad CBS contract tariffs and component bills documented by [de Bondt et al. \(2024\)](#), adding the cap discount back at the same reference quantities while retaining fixed charges and taxes. The Dutch calculation is a conditional benchmark rather than an observed all-household counterfactual because contract shares and household consumption distributions are unavailable ([Haan and Schinkel, 2023](#)).

Subsidies, Rebates, and Other Fuel-Price Reductions. For unit subsidies and fuel rebates, we add the statutory absolute price wedge to the observed monthly price path or, without a retail unit-price series, calibrate it in HICP component-index units. Evidence from [Drolsbach et al. \(2023\)](#) suggests that temporary fuel-tax reductions and discounts were largely passed through to consumer prices in 2022 in several euro-area countries, although pass-through varied over time and by fuel type. This supports treating these measures as retail-price wedges in the counterfactual construction. Under full pass-through, adding the statutory support amount $s_{j,t}$ to the observed unit price gives:

$$\tilde{r}_{j,t} = r_{j,t} + s_{j,t}. \quad (12)$$

Germany's 2023 price brakes provided bill rebates on 80 percent of a predetermined reference quantity ([Dertwinkel-Kalt and Wey, 2025](#), [Krebs and Weber, 2024](#)). We use a representative-bill model anchored on Bundesnetzagentur's April 2023 prices and infer the uncapped bill from monthly HICP movements under fixed-quantity assumptions. This is a conditional benchmark because household contract shares and consumption distributions are unavailable. Spain's transport-fuel rebates are reconstructed from the statutory unit discounts. For transport-fuel rebates, $s_{j,t}$ is recovered from the statutory pump discount, expressed in euros per litre, and applied to a monthly retail-fuel price $r_{j,t}^{\text{M}}$, where the upright superscript M denotes the observed market price per litre corresponding to HICP component j . Following [Bonnet et al. \(2025\)](#), we refer to these interventions as pump-price rebates or subsidies because they reduced the tax-inclusive price paid by consumers, although their

technical implementation sometimes operated through fuel excise taxation rather than through a direct transfer to households. The no-rebate fuel index is obtained by rescaling the observed HICP transport-fuel component:

$$\tilde{p}_{j,t} = p_{j,t} \frac{r_{j,t}^M + s_{j,t}}{r_{j,t}^M}.$$

These transport-fuel measures are applied only to fuels for personal transport equipment. The product decompositions nevertheless report the broader vehicle-operation aggregate; its remaining components retain their observed price paths unless they are affected by a separately identified measure. When the policy inventory identifies a subsidy but does not supply a monthly retail unit-price series, we express its constant absolute price wedge in component-index units. For measure k affecting product j , let $\bar{p}_j$ denote the arithmetic mean of the observed component index over that measure's active months. We set $s_{k,t} = \delta_k \bar{p}_j$ during those months, yielding the monthly counterfactual-to-observed ratio $1 + \delta_k \bar{p}_j / p_{j,t}$. We choose δ_k so that the geometric mean of this ratio over the active months equals the measure's assumed component-level price ratio.

Other Measures Requiring Calibration. For a small set of price-relevant interventions, the policy inventories identify the measure but do not provide the monthly counterfactual tariff schedule, the distribution of household consumption across tariff blocks, or the eligible-household weights needed for a direct statutory reconstruction. We therefore add these cases as separate calibrated layers, identified from [Sgaravatti et al. \(2021\)](#) and [Arregui et al. \(2022\)](#), rather than merging them with the VAT or unit-subsidy calculations. This applies to Cyprus's electricity-bill subsidy, Ireland's electricity credits and Public Service Obligation levy reduction, Italy's system-charge reductions and social energy bonuses, and Belgium's extension of the social energy tariff and basic energy package. The calibrated ratios are applied only to the relevant COICOP energy components and policy months, and the corresponding assumption tables are reported with the counterfactual outputs.

We then replace observed component indices with these counterfactual paths and apply the same RAS-calibrated weights and household-group, national, and euro-area aggregation procedures as in [Section 1](#).

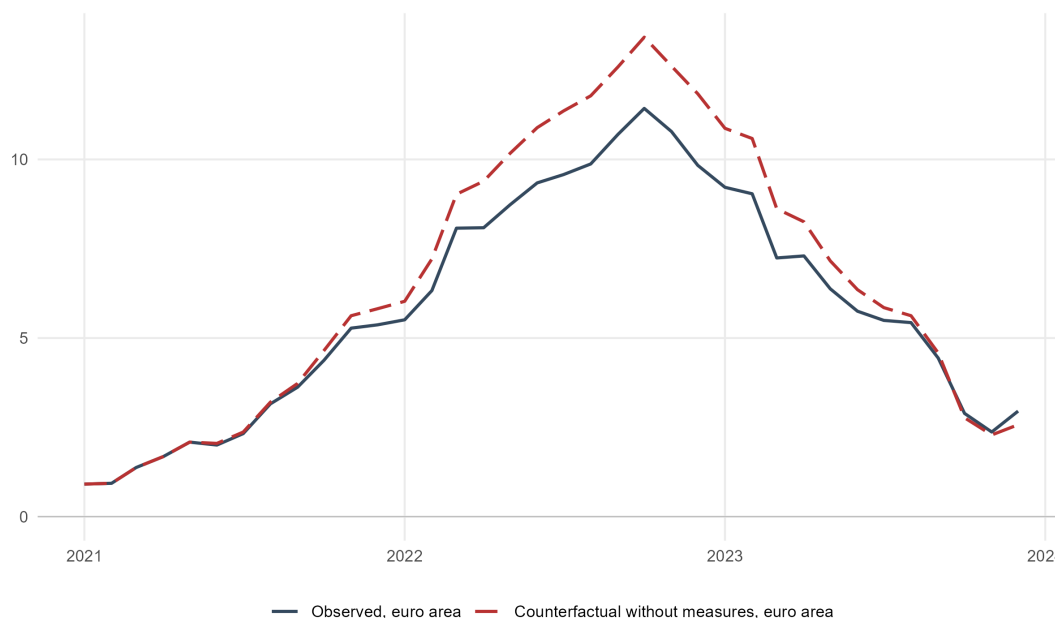
3.2 Euro-Area Effects on Inflation and Inflation Inequality

The price-based measures changed both the level and the timing of inflation during the energy crisis. We isolate their direct effect on consumer prices. Observed minus counterfactual euro-area

annual inflation is -0.1 percentage point in 2021, -1.5 points in 2022, and -0.6 point in 2023 (Table 7). The 2022 estimate is consistent in magnitude with [Dao et al. \(2023\)](#), who find that unconventional fiscal policy measures reduced euro-area inflation by 1–2 percentage points in 2022 under different modeling assumptions. Figure 7 is constructed from the same observed and counterfactual product indices as Table 7; annual averaging of the plotted indices therefore reproduces the table estimates. Counterfactual inflation lies above observed inflation during the main phase of the energy shock, with the largest difference around the inflation peak in late 2022. VAT and non-fuel excise reductions, fuel-price reductions, price shields, and regulated-price caps therefore compressed the peak.

A reduction in the price level need not lower inflation in every subsequent year. When a temporary measure expires or its price discount shrinks, observed prices rise relative to the lower comparison base created by the measure. Observed annual inflation can then exceed counterfactual inflation even if the observed price level remains no higher than its no-policy counterpart. Positive annual policy effects therefore can reflect the withdrawal of earlier support and base effects, rather than a higher price level caused by the policy. The same timing mechanism can reverse a policy's annual effect on the Q1–Q5 inflation gap.

Figure 7: Household price policies: observed and counterfactual euro-area inflation (year on year, in %)

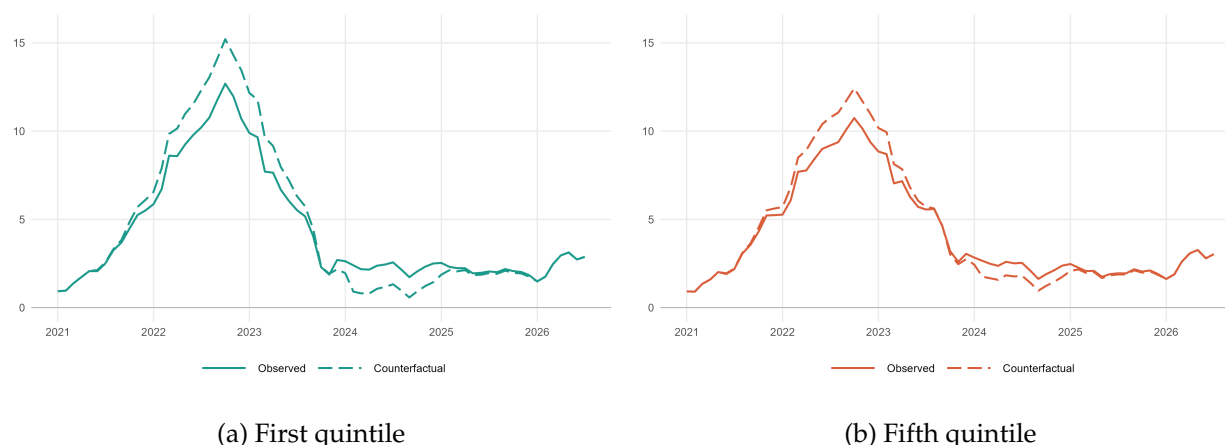


Notes: The counterfactual series removes fiscal interventions that directly affected consumer prices. The figure reports year-on-year inflation for all 20 euro-area countries, aggregated with annual HICP country weights, from January 2021 to December 2023. Policy effects are defined as observed minus counterfactual inflation. The plotted indices are the same product-to-country-to-euro-area indices used for the annual estimates in Table 7.

Sources: Eurostat HICP; Bruegel energy-policy dataset; authors' calculations.

Figure 8 compares observed and counterfactual inflation separately for the first and fifth income quintiles. The reduction around the 2022 peak is larger for the first quintile, so the policies narrow the Q1–Q5 gap during the energy shock. This pattern indicates greater price relief for the Q1 consumption basket when energy prices were rising fastest. It reflects differences in basket exposure to the covered price measures and does not identify household eligibility or the benefits actually received. This finding is consistent with microsimulation evidence that the European fiscal response reduced distributional losses, although broad price measures were less targeted than income support (Amores et al., 2025).

Figure 8: Household price policies: observed and counterfactual euro-area inflation by income quintile (year on year, in %)



Notes: Solid lines show observed year-on-year inflation and dashed lines show inflation in the absence of household price policies for the first and fifth income quintiles, from January 2021 to July 2026. Income transfers and firm-only measures are outside the counterfactual. Both scenarios use the same group baskets constructed primarily from individual 2020 HBS records, with the Italian approximation and Portuguese tabulation fallback. All 20 countries use RAS expenditure calibration for these income groups and are aggregated using annual HICP country weights.

Sources: Eurostat HICP and country weights; household budget surveys; Bruegel energy-policy dataset; authors' calculations.

Table 4 decomposes these annual policy effects by instrument and targeting status and reports the corresponding documented fiscal costs. The fiscal accounting assigns a cost to each measure active in the counterfactual during 2021–2023 and counts shared envelopes once, yielding an EA20 total of 1.58 percent of 2022 GDP. Some multi-instrument envelopes cannot be allocated comparably across instruments or targeting classes. The table therefore reports a dash for the affected category-level costs while retaining these envelopes in the total. Some envelopes include business beneficiaries, so these costs do not isolate amounts received by households.

Administered-price measures reduce the Q1–Q5 inflation gap by 0.4 point in both 2022 and 2023. Targeted measures reduce it by 0.3 point in 2022 and 0.1 point in 2023; untargeted measures reduce it by 0.3 and 0.5 points, respectively.

The fuel-price-reduction category includes all road-fuel (CP0722) measures, including excise-duty reductions, regulated-price support, and direct subsidies. The VAT and non-fuel excise category therefore excludes road-fuel measures and covers VAT reductions and other non-fuel tax and charge relief. The other calibrated layer mainly comprises Italian social energy bonuses and electricity-system charge relief. Fiscal costs and annual inflation effects are different objects: the

latter depend on the affected products, policy timing and price assumptions, and are not calibrated to the fiscal amounts.

Table 4: Annual effects of household price-policy categories on average inflation and inflation inequality in the euro area, 2021–2023

		Effect of price policies on average inflation			Effect of price policies on Q1–Q5 inflation inequality		
	Fiscal cost (% of 2022 GDP)						
Policy category		2021	2022	2023	2021	2022	2023
<i>Instrument classification</i>							
VAT and non-fuel excise cuts	–	-0.1	-0.4	-0.2	0.0	-0.2	-0.1
Administered prices	–	0.0	-0.7	-0.7	0.0	-0.4	-0.4
Fuel price reductions	–	0.0	-0.3	0.3	0.0	0.1	0.0
Other calibrated layer	–	0.0	-0.1	0.0	0.0	0.0	0.0
Total	1.58	-0.1	-1.5	-0.6	-0.1	-0.6	-0.5
<i>Targeted vs Non-targeted</i>							
Untargeted price measures	–	-0.1	-1.0	-0.4	0.0	-0.3	-0.5
Targeted price measures	–	0.0	-0.5	-0.2	0.0	-0.3	-0.1
Total	1.58	-0.1	-1.5	-0.6	-0.1	-0.6	-0.5

Notes. Observed minus counterfactual annual inflation or Q1–Q5 gap, in percentage points; negative values indicate a reduction. Fuel price reductions include all road-fuel (CP0722) relief, including excise-duty reductions, regulated-price support, and direct subsidies; the VAT and non-fuel excise category excludes road-fuel measures. Category effects need not sum to Total because policies interact. Targeting classifies measures, not individual eligibility. Fiscal costs are recorded policy envelopes divided by nominal 2022 GDP for the same 20 countries; budget periods may span several years. Fiscal costs refer to entire schemes affecting household prices, not amounts received exclusively by households. A dash indicates an unidentified comparable cost, not zero. Sources: Eurostat, national sources, authors' calculations.

Table 5 separates the role of the three construction methods. Keeping only direct statutory or published-price reconstructions gives average-inflation effects of 0.0, –0.8, and 0.0 points in 2021–2023, compared with –0.1, –1.5, and –0.6 points in the full specification. The corresponding Q1–Q5 effects are 0.0, –0.3, and –0.1 points, so the conclusion that the policies narrowed the gap survives without the less directly observed layers. Removing only the conditional bill and tariff benchmarks changes the 2023 average-inflation effect from –0.6 to –0.2 point and the gap effect from –0.5 to –0.3 point. Scaling those conditional wedges to 50 or 150 percent yields 2023 effects between –0.4 and –0.8 point for average inflation and between –0.4 and –0.7 point for the gap. These differences measure dependence on counterfactual assumptions; they are not confidence intervals.

Table 5: Sensitivity of euro-area inflation effects to counterfactual construction methods

Scenario	Average inflation effect			Q1–Q5 gap effect		
	2021	2022	2023	2021	2022	2023
Full specification	-0.1	-1.5	-0.6	-0.1	-0.6	-0.5
Direct reconstructions only	0.0	-0.8	0.0	0.0	-0.3	-0.1
Full specification without conditional benchmarks	-0.1	-1.5	-0.2	-0.1	-0.6	-0.3
Conditional benchmark wedges at 50%	-0.1	-1.5	-0.4	-0.1	-0.6	-0.4
Conditional benchmark wedges at 150%	-0.1	-1.5	-0.8	-0.1	-0.6	-0.7
Dependence on estimated layers: full minus direct-only	-0.1	-0.7	-0.5	0.0	-0.3	-0.4

Notes. Entries are observed minus counterfactual annual inflation or Q1–Q5 inflation gaps, in percentage points. Direct reconstructions use statutory rates, unit rebates, or published reference-price paths without a fitted intensity. The full specification also includes calibrated paths and conditional German and Slovenian bill or tariff benchmarks. The 50% and 150% rows scale only the price wedges of those conditional benchmarks. Full minus direct-only measures dependence on estimated layers; it is not a confidence interval. Sources: Eurostat, national sources, authors' calculations.

3.3 Heterogeneity Across Products and Countries

We first identify the products through which household price policies affected inflation inequality, then examine how the resulting effects differed across national policy packages.

3.3.1 Differences Across Products

Table 6 decomposes the aggregate policy effects reported above into product-group contributions to average inflation and the Quintile 1–Quintile 5 gap.

Table 6: Product-group contributions of household price policies to annual-average euro-area inflation and Q1–Q5 inflation inequality, 2021–2023

Product group	Weight (2022, %)	Effect of price policies on average inflation			Effect of price policies on Q1–Q5 inflation inequality		
		2021	2022	2023	2021	2022	2023
Food and non-alcoholic beverages	17.8	0.0	0.0	-0.1	0.0	0.0	0.0
Alcoholic beverage, tobacco and narcotics	4.6	0.0	0.0	0.0	0.0	0.0	0.0
Clothing and footwear	5.6	0.0	0.0	0.0	0.0	0.0	0.0
Housing maintenance and furnishings	8.0	0.0	0.0	0.0	0.0	0.0	0.0
Water, electricity, gas and other fuels	6.8	-0.1	-1.2	-0.8	-0.1	-0.6	-0.5
Operation of personal transport equipment	6.3	0.0	-0.3	0.3	0.0	0.1	0.0
Transport (excluding vehicle operation)	9.6	0.0	0.0	0.0	0.0	0.0	0.0
Other	41.1	0.0	0.0	0.0	0.0	0.0	0.0
Total	100.0	-0.1	-1.5	-0.6	-0.1	-0.6	-0.5

Notes. Policy effects are observed minus counterfactual product contributions, in percentage points; the gap is Q1 minus Q5. Annual columns compare annual-average indices. Weights are 2022 EA20 expenditure shares. Sources: Eurostat, national statistical institutes, authors' calculations.

Housing energy is the main contributor to the reduction in the inflation gap: -0.1 point in 2021, -0.6 point in 2022, and -0.5 point in 2023. Electricity and gas receive larger budget weights among lower-income households, so relief on these prices narrows the Q1–Q5 gap. Vehicle-operation expenditure partly offsets this reduction, contributing 0.1 point to the gap in 2022 and rounding to zero in 2023; within this group, the policy wedge applies only to fuels and lubricants. Its effect on average inflation reverses from -0.3 point in 2022 to 0.3 point in 2023 as earlier relief unwinds, partly offsetting the 0.8-point reduction from household energy. Food policies contribute a further 0.1-point reduction in average inflation in 2023. Thus, the distributional effect depends on both the affected product and the change in support relative to the previous year; a policy's effect on average inflation alone does not determine its effect on inflation inequality. Product contributions are rounded separately, so displayed rows need not sum exactly to the displayed total. We next examine how these product-level effects combine within national policy packages.

3.3.2 Differences Across Countries

Table 7 reports the annual effects of each country's household price policies on average inflation and the Quintile 1–Quintile 5 inflation gap, together with their documented fiscal costs. The costs describe the scale of the schemes rather than household incidence; an asterisk identifies envelopes

that also include business beneficiaries.

Table 7: Household price policies: fiscal cost and effects on annual-average inflation and Q1–Q5 inflation inequality in the 20 euro-area countries, 2021–2023

Country	Weight (2022, %)	Fiscal cost (% of 2022 GDP)	Effect of price policies on average inflation			Effect of price policies on Q1–Q5 inflation inequality		
			2021	2022	2023	2021	2022	2023
Germany	28.1	1.2*	0.0	-0.7	-1.5	0.0	-0.3	-1.2
France	20.4	2.9	-0.1	-2.7	-0.6	0.0	-1.1	-0.5
Italy	16.3	1.5	-0.2	-1.4	0.7	-0.1	-0.2	-0.1
Spain	11.0	2.0*	-0.5	-2.1	-0.3	-0.2	-0.8	-0.3
Netherlands	5.4	1.0*	0.0	-2.5	-0.5	0.0	-1.2	-0.3
Belgium	4.0	0.7*	0.0	-2.4	0.5	0.0	-1.0	0.1
Austria	3.3	0.9*	0.0	-0.2	-1.0	0.0	-0.1	-0.5
Portugal	2.2	1.5	0.0	-1.0	-1.8	0.0	-0.4	-1.1
Greece	2.2	2.4*	-0.5	-1.6	0.3	-0.4	-0.8	-0.1
Finland	1.9	0.2	0.0	-0.1	-0.3	0.0	0.1	0.0
Ireland	1.5	0.4*	0.0	-0.8	0.4	0.0	-0.5	0.1
Slovakia	0.8	1.7	0.0	0.0	-3.1	0.0	0.0	-1.7
Croatia	0.7	0.9	0.0	-1.1	0.3	0.0	-0.9	0.4
Lithuania	0.6	0.8	0.0	-0.6	-0.1	0.0	-0.5	-0.1
Slovenia	0.4	0.3	0.0	-0.8	0.0	0.0	-0.1	-0.5
Luxembourg	0.3	0.8*	0.0	-0.4	-0.6	0.0	-0.1	-0.2
Latvia	0.3	1.1	0.0	-1.1	0.1	0.0	-1.3	0.2
Estonia	0.2	0.5*	-0.2	-0.8	0.3	-0.1	-0.6	0.3
Cyprus	0.2	0.7	-0.1	-0.7	0.6	0.0	-0.3	0.2
Malta	0.1	2.6*	0.0	-1.4	0.0	0.0	-0.3	0.0
Euro area	100.0	1.6	-0.1	-1.5	-0.6	-0.1	-0.6	-0.5

Notes. Effects are observed minus counterfactual annual inflation, in percentage points; the gap is Q1 minus Q5. Costs are scheme budgets as a percentage of 2022 GDP, counting shared envelopes once. An asterisk marks costs including business beneficiaries; the household share is not identified. A dash denotes incomplete coverage, not zero. France's fuel rebate excludes EUR 1.6 billion paid to road freight; its electricity shield also covers microbusinesses. Source periods vary (France and Lithuania: 2022); Slovak amounts are gross 2023 expenditure. Dutch quintile effects measure basket exposure, not cap eligibility. Country weights are 2022 HICP shares. Sources: Eurostat, Bruegel, national sources, authors' calculations.

The distributional effects are heterogeneous across countries. In 2022, the largest reductions in the Q1–Q5 gap occur in Latvia (1.3 points), the Netherlands (1.2), France (1.1), Belgium (1.0), and Croatia (0.9). In 2023, Slovakia records the largest reduction (1.7 points), followed by Germany (1.2), Portugal (1.1), and France (0.5). Policies increase the gap by 0.4 point in Croatia, 0.3 in Estonia, and 0.2 in Cyprus in 2023. Italy’s gap effects are -0.1 , -0.2 , and -0.1 points in 2021–2023, while its effect on average inflation turns positive, at 0.7 point, in 2023.

For Germany, annual-average inflation is reduced by 0.7 point in 2022 and 1.5 points in 2023, while the Q1–Q5 gap is reduced by 0.3 and 1.2 points. The direction of this accounting result is consistent with [Krebs and Weber \(2024\)](#), who argue that the German price brakes lowered measured inflation and supported household purchasing power. The estimate does not include the endogenous supplier-pricing and energy-saving responses analyzed by [Dertwinkel-Kalt and Wey \(2025\)](#).

The Netherlands also records a large reduction in the gap in 2022, but this estimate captures quintile-specific basket exposure to common component-price paths rather than household eligibility for the partially covered cap or the distribution of cap transfers, a distinction emphasized by [Haan and Schinkel \(2023\)](#). For France, our direct accounting exercise yields reductions in annual-average inflation of 2.7 points in 2022 and 0.6 point in 2023; the corresponding reductions in the Q1–Q5 gap are 1.1 and 0.5 points. Using a HANK model, [Langot et al. \(2026\)](#) estimate that the French tariff shield lowers inflation from 7.2 to 6.5 percent in 2022 and from 3.6 to 3.5 percent in 2023. Their smaller net effects reflect a different object: their model incorporates general-equilibrium demand and wage-price feedback, whereas our estimates isolate the direct effect of a broader set of household price policies on consumer prices. France’s documented active-measure costs amount to 2.9 percent of 2022 GDP. This price relief should not be read as a welfare or targeting result: for French fuel-price support, [Bonnet et al. \(2025\)](#) find that income-targeted transfers limit welfare losses relative to income more effectively than price subsidies. Slovakia’s recorded gross expenditure attributed to 2023 amounts to 1.7 percent of 2022 GDP. Differences in budget periods or forecast and actual amounts prevent these figures from supporting a fiscal-efficiency ranking. A cost-effectiveness comparison would require eligibility, quantities consumed, tax and transfer rules, and counterfactual expenditures on a common monetary and beneficiary basis. Inflation-gap effects are percentage-point changes, not monetary household gains. Taken together, the country results show that policy effects on inflation inequality arise from the interaction between policy design and household expenditure shares. Fiscal cost alone therefore does not predict the size or direction of the Q1–Q5 inflation-gap effect.

4 Household Inflation Exposure: Shocks and Burden

This section assesses direct and indirect exposure to extra-EU import-price shocks and measures the inflation burden relative to household income.

4.1 Direct and Indirect Exposure to Extra-EU Import-Price Shocks

We apply the observed extra-EU import-price changes to imported final goods and intermediate inputs. The FIGARO production network propagates the intermediate-input cost impulses across EA20 countries and industries. We then map producer-price responses into consumption products and aggregate them with the household expenditure weights used elsewhere in the paper. Throughout this subsection, $k \in \{\text{energy, food}\}$ indexes the import-price shock, d denotes the destination country, i an imported product, j a destination industry, and g a household group. A superscript k labels the shock scenario; it is not an exponent.

Let h_k denote the observed percentage change in the annual-average extra-EU import unit-value index for shock k . For a final product i absorbed by households in country d , let F_{di}^{extra} denote imports from outside the EU27 and F_{di}^{all} total household absorption, including domestic production and all imports. The direct final-demand price impulse is

$$d_{di}^k = h_k \frac{F_{di}^{\text{extra}}}{F_{di}^{\text{all}}}. \quad (13)$$

The ratio therefore scales the common external price shock by the extra-EU share of the product's household supply. The vector $\mathbf{d}^k$ stacks d_{di}^k over all country-product pairs.

For intermediate inputs, $\mathbf{A}$ is the FIGARO matrix of input coefficients for the EA20 country-industry network: each element records the amount of an input required per unit of gross output in the purchasing industry. The vector $\mathbf{b}^k$ contains the first-round production-cost impulses generated by extra-EU inputs affected by shock k . Its element for industry j in country d is

$$b_{dj}^k = h_k \frac{\sum_{o \notin \text{EU27}} \sum_{i \in S_k} Z_{oi,dj}}{x_{dj}}, \quad (14)$$

where o indexes the origin country, S_k is the set of products affected by shock k , $Z_{oi,dj}$ is the FIGARO intermediate flow of product i from origin o to industry j in destination country d , and x_{dj} is that industry's gross output. Thus, b_{dj}^k is the first-round increase in unit production costs under complete pass-through of the import-price shock. The vector $\mathbf{b}^k$ stacks these impulses over all EA20 country-industry pairs.

The corresponding producer-price response is

$$\Delta \mathbf{p}^k = (\mathbf{I} - \mathbf{A}^\top)^{-1} \mathbf{b}^k, \quad (15)$$

where $\mathbf{I}$ is the identity matrix and $(\mathbf{I} - \mathbf{A}^\top)^{-1}$ is the Leontief price inverse. The vector $\Delta \mathbf{p}^k$ includes both the initial imported-input cost impulse and all subsequent rounds of cost propagation through domestic and intra-EA20 input links.

Finally, $\mathbf{C}$ is the normalized NACE–COICOP concordance matrix: multiplying an industry-level price vector by $\mathbf{C}$ yields the corresponding vector of COICOP consumption-price changes. The vector $\mathbf{w}_g$ contains the normalized COICOP expenditure shares of household group g , so its elements sum to one. The total inflation exposure of group g is

$$\Delta \pi_g^k = \mathbf{w}_g^\top \mathbf{C} \mathbf{d}^k + \mathbf{w}_g^\top \mathbf{C} \Delta \mathbf{p}^k. \quad (16)$$

The first term is the contribution of directly imported final goods; the second is the contribution of imported intermediate inputs after production-network propagation. Because h_k and all price responses are expressed as percentage changes, $\Delta \pi_g^k$ is an inflation exposure in percentage points. The effect on inflation inequality is $\Delta \pi_{Q1-Q5}^k = \Delta \pi_{Q1}^k - \Delta \pi_{Q5}^k$, where a positive value means that shock k raises the inflation exposure of the first income quintile more than that of the fifth.

Panel (a) uses the observed increase in the annual-average extra-EU energy import unit-value index from 2021 to 2022, and panel (b) uses the corresponding food import-price increase. The raw shocks, their conversion into country-industry cost impulses, and the propagation equations are reported in Appendix B. The shocks are common external-price innovations, while bilateral import shares, production structures, and household baskets generate heterogeneous country and household exposure.

Table 8: Product-group contributions of observed extra-EU import-price shocks, euro area, 2022

Product group	Average inflation			Q1–Q5 gap		
	Total	Direct	Indirect	Total	Direct	Indirect
<i>(a) Energy import-price shock: 83.9%</i>						
Food and non-alcoholic beverages	0.6	0.0	0.6	0.2	0.0	0.2
Alcoholic beverages, tobacco and narcotics	0.1	0.0	0.1	0.0	0.0	0.0
Clothing and footwear	0.1	0.0	0.1	0.0	0.0	0.0
Housing maintenance and furnishings	0.2	0.0	0.2	-0.1	0.0	-0.1
Water, electricity, gas and other fuels	1.7	0.5	1.2	1.7	0.5	1.2
Operation of personal transport equipment	1.8	0.3	1.5	-0.5	-0.1	-0.5
Transport (excluding vehicle operation)	1.2	0.2	1.0	0.0	0.0	0.0
Other	0.8	0.2	0.5	-0.1	0.0	-0.1
Total	6.5	1.3	5.2	1.1	0.4	0.7
<i>(b) Food import-price shock: 21.0%</i>						
Food and non-alcoholic beverages	0.5	0.3	0.2	0.2	0.1	0.1
Alcoholic beverages, tobacco and narcotics	0.1	0.1	0.0	0.0	0.0	0.0
Clothing and footwear	0.0	0.0	0.0	0.0	0.0	0.0
Housing maintenance and furnishings	0.0	0.0	0.0	0.0	0.0	0.0
Water, electricity, gas and other fuels	0.0	0.0	0.0	0.0	0.0	0.0
Operation of personal transport equipment	0.0	0.0	0.0	0.0	0.0	0.0
Transport (excluding vehicle operation)	0.0	0.0	0.0	0.0	0.0	0.0
Other	0.0	0.0	0.0	0.0	0.0	0.0
Total	0.7	0.4	0.2	0.2	0.1	0.1

Notes. Shocks are changes in the 2022 annual-average extra-EU27 import unit-value indices relative to 2021. Energy is SITC 3; food is SITC 0–1. Direct effects use extra-EU imports purchased by households. Indirect effects use extra-EU intermediate inputs and the full EA20 FIGARO propagation. Direct and indirect effects sum to the total before rounding.

Sources: Eurostat international trade statistics, HBS, FIGARO, authors' calculations.

Table 8 shows that the observed energy import-price shock generates 6.5 percentage points of average EA20 non-rent inflation exposure: 1.3 points through imported final goods and 5.2 points through intermediate inputs and production-network propagation. Household energy and vehicle operation contribute 1.7 and 1.8 points, respectively. The net Q1–Q5 effect is 1.1 points, comprising 0.4 point directly and 0.7 point indirectly. Household energy contributes 1.7 points to the gap, partly offset by vehicle operation (–0.5 point), housing maintenance and other products. The food import-price shock generates 0.7 point of average exposure and a Q1–Q5 gap of 0.2 point. Components and totals are rounded separately.

These are accounting exposures under complete pass-through, not causal estimates or a model fit to observed HICP inflation. Retail margins, taxes, subsidies, regulated prices, substitution, and incomplete pass-through can all attenuate or redistribute the observed consumer-price response.

4.2 Inflation Burden

Our indicators measure differences in inflation exposure across household groups, but they do not directly express the cost of inflation relative to household resources. We therefore construct the burden from HBS microdata rather than combining group-level inflation with an external aggregate consumption-to-income ratio. For household i in country c , let X_{ic} denote consumption expenditure on the non-rent products matched to HICP, Y_{ic} net household income, and $\bar{\pi}_{ic,y}$ the household's average annual inflation rate in year y , computed with the RAS-calibrated weights described in Section 1.2. The raw household-level burden is

$$\mathcal{B}_{ic,y}^{\text{raw}} = \bar{\pi}_{ic,y} \frac{X_{ic}}{Y_{ic}}, \quad (17)$$

where $\bar{\pi}_{ic,y}$ is measured in percent, so $\mathcal{B}_{ic,y}^{\text{raw}}$ is measured in points of net income. Thus, the indicator allows both individual inflation exposure and the consumption-to-income ratio to vary across households. The expenditure numerator and inflation index cover the same matched non-rent products. This fixed-resource exposure indicator is not compensating variation: it holds 2020 expenditure and income fixed and allows neither substitution nor income adjustment.

Very small reported incomes can generate mechanically extreme expenditure-to-income ratios. We therefore winsorize $100X_{ic}/Y_{ic}$ at the survey-weighted 99th percentile within each country, while retaining the underlying household inflation rate. Denoting the resulting ratio by $\tilde{\kappa}_{ic}$, the burden used in the estimates and its aggregation for national income quintile q are

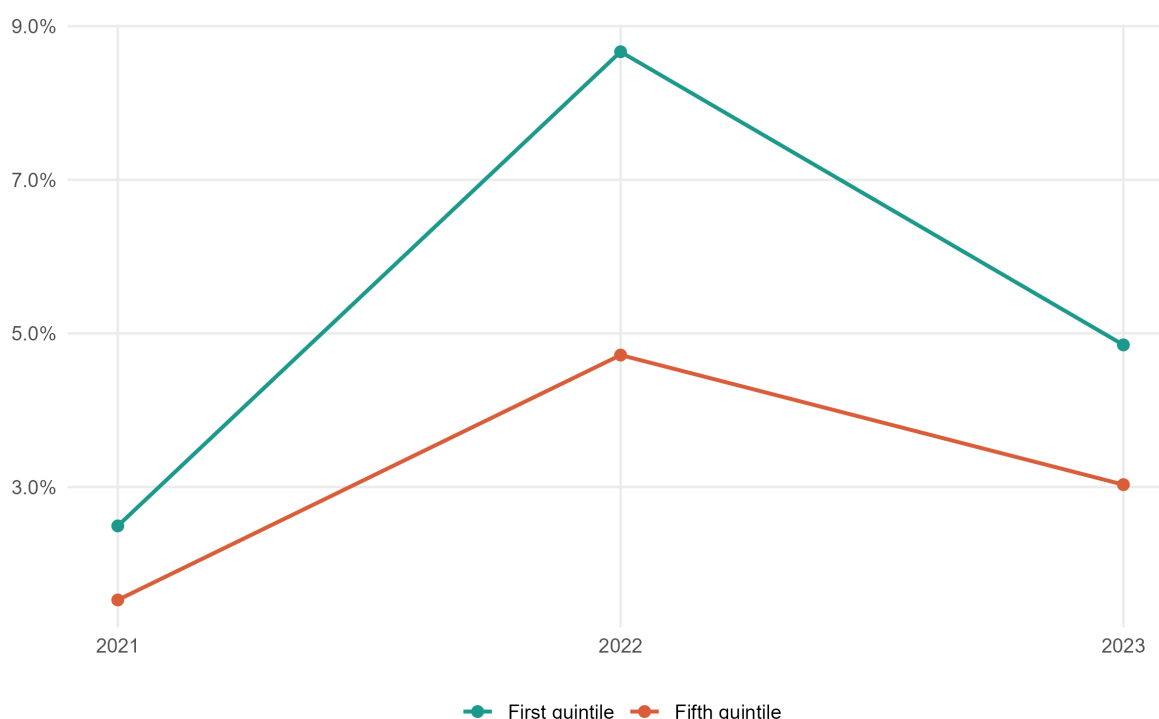
$$\mathcal{B}_{ic,y} = \bar{\pi}_{ic,y} \frac{\tilde{\kappa}_{ic}}{100}, \quad \mathcal{B}_{q,y} = \frac{\sum_c \sum_{i \in q(c)} a_{ic} \mathcal{B}_{ic,y}}{\sum_c \sum_{i \in q(c)} a_{ic}}, \quad (18)$$

where a_{ic} is the HBS survey weight and $q(c)$ indicates that total-net-income quintiles are defined separately within each country before households are pooled. The survey weights expand the microdata to national household populations and therefore determine the euro-area aggregation. The 2020 HBS basket, income, and expenditure observations are held fixed, while household-specific inflation is calculated for 2021–2023. For each household-year, the inflation rate is computed over the matched non-rent expenditure categories and the HBS–HICP product weights are renormalized to one. We impose no minimum matched-expenditure threshold.

Higher-income households also generally have greater capacity to absorb a price shock by reducing discretionary expenditure or substituting toward cheaper varieties. Appendix Figure C.2 documents the corresponding difference in basket composition.

Figure 9 uses household indices and matched non-rent expenditure. The first quintile faces a burden of 2.5 percent of net income in 2021, 8.7 percent in 2022, and 4.8 percent in 2023, compared with 1.5, 4.7, and 3.0 percent for the fifth quintile. The Q1–Q5 gap is 1.0, 4.0, and 1.8 points, respectively, calculated before rounding. The income and expenditure filters retain positive-income households; national quintiles are assigned before these burden-specific exclusions.

Figure 9: Household-level inflation burden in the euro area, by income quintile (in % of net income)



Notes: Annual household inflation times matched non-rent expenditure divided by net income; ratios are winsorized at the national survey-weighted 99th percentile. Survey-weighted means within national total-income quintiles, 18 countries excluding Italy and Portugal. Fixed 2020 resources; this is an exposure measure, not a welfare estimate.

Sources: HBS microdata, Eurostat HICP, authors' calculations.

5 Conclusion

Household expenditure baskets reveal differences in exposure to the recent inflation surge. Modest income-quintile gaps at the euro-area level conceal larger differences within several countries. Older and rural households also experience higher inflation during the rise in energy prices, although these associations vary over time.

Differences in household resources amplify the burden of inflation. Our fixed-resource exposure measure is higher relative to net income for households in the lowest income quintile than for those in the highest. Similar inflation rates can therefore impose different burdens relative to income.

The price-policy counterfactual indicates that household price measures lower average inflation during the energy crisis and narrow the gap between the lowest and highest income quintiles. Relief on household energy prices accounts for most of the narrowing, while motor-fuel measures partly offset it. These accounting effects depend on the reconstructed and calibrated price wedges and do not measure the policies' full welfare or macroeconomic effects.

A model saturated with all interactions among observed household characteristics explains only 12.4% of household inflation variance on average. The remaining 87.6% lies within observationally similar groups. The main distributional gradients are therefore informative about systematic exposure, but they summarize only a limited part of household inflation dispersion.

Further work could use transaction data to identify differences in prices paid and substitution within product categories, while linked income and balance-sheet data would permit a broader assessment of welfare effects, including the redistribution between debtors and creditors.

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Appendix

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(Not for publication)

October 7, 2026

A Aggregation Bias Across COICOP Digits

Table [A.1](#) compares inflation gaps across COICOP digit depths using HBS microdata and reports a digit-3 benchmark from Eurostat national-quintile HBS aggregates. Macro D3 excl. and all microdata columns exclude actual and imputed rents (HBS 041 and 042) and the corresponding HICP actual-rent component (041), and use RAS-calibrated weights. All columns use annual-average chained-index inflation. The macro benchmark is recalculated with RAS; the rent-robustness table retains its relative-expenditure specification.

Micro–macro consistency. We first reaggregate the 2020 household records into national income quintiles and compare survey-weighted expenditure shares with Eurostat’s published HBS shares on a common digit-3 support, excluding rents. Ranking households by total net household income substantially improves this match relative to ranking them by equivalised income. Across products and quintiles, the mean absolute difference in budget shares falls from 0.20 to 0.05 percentage points in the Netherlands and from 0.31 to 0.04 in Belgium; the corresponding figures for France are 0.19 and 0.03. This comparison precedes HICP calibration and therefore identifies a difference in the construction of income groups rather than an effect of the inflation-index weighting method. We use total household income for the microdata quintiles throughout this appendix.

This appendix retains its original comparison design: expenditure is rebuilt from the finest available codes and quintile calibration margins come from published tabulations. The main microdata pipeline instead prioritizes directly observed expenditure at the required depth and computes group expenditure margins from the same household sample. The appendix therefore provides a separate digit-depth sensitivity exercise, rather than a replication of the updated baseline.

Baskets and weights. Microdata quintiles are defined separately within each country using survey-weighted net household income without division by an equivalence scale (total household income). The sample covers 18 euro-area countries. Fifteen countries provide expenditure variables through digit 5; Germany, France and Spain provide digit 3 in the local microdata files. Italy is excluded because household income is unavailable, and Portugal because its detailed expenditure columns are unpopulated. The 2020 microdata expenditure baskets are reconstructed at each digit depth from the same finest available expenditure variables after removing rents. Nonpositive or missing expenditure entries contribute zero, while eligible households remain in the survey denominators.

For product j in quintile q , let $\bar{x}_{qj}$ be survey-weighted mean HBS expenditure, $\bar{x}_j$ the national mean, $s_{q,y}$ the package's consumption share of quintile q , and $w_{j,y}^{HICP}$ the HICP product share on matched support. RAS iteratively rescales rows and columns of the positive seed

$$\max\{s_{q,y}w_{j,y}^{HICP}\bar{x}_{qj}/\bar{x}_j, 10^{-12}\}.$$

The calibrated expenditure masses $M_{qj,y}$ satisfy

$$\sum_j M_{qj,y} = s_{q,y}, \quad \sum_q M_{qj,y} = w_{j,y}^{HICP}, \quad \omega_{qj,y} = M_{qj,y}/s_{q,y}. \quad (19)$$

Both the macro and micro calculations use the package's published consumption shares by income quintile, selecting the latest available wave preceding each weight year, and annual HICP product margins. These quintile margins are not re-estimated from the matched microdata at each digit depth. A shared ECOICOP1 correspondence table defines the D2–D5 mappings for both the household-index function and this appendix. Expenditures and historical HICP prices use the same canonical product codes; rents are removed before D2 aggregation. Micro expenditures are aggregated by quintile before RAS calibration; the table does not average separately chained household indices. Macro baskets use the corresponding published Eurostat HBS expenditures, selecting the most recent available preceding wave for each matched item. Micro prices use the archived ECOICOP1 HICP indices and item weights, which match the HBS classification, and are reaggregated with annual HICP weights. An undivided terminal HICP class is matched to its identical HBS code padded with zeros (for example, electricity 0451/04510 and hospital services 063/06300). Five additional terminal correspondences for hospital services and education are validated against the HBS manual and Eurostat labels and recorded in an explicit mapping table. Each corresponding published price and weight enters once; no distinct subclass receives an imputed parent price. Fuzzy label matching is used to propose candidates, and unvalidated candidates are excluded. Rent components are removed before constructing the broad housing

index. Products without complete monthly prices or annual weights are excluded; prices are not imputed to distinct subclasses. Matched expenditure coverage can vary across digit depths, so the reported differences can combine aggregation and coverage effects. The macro benchmark retains the price source and classification bridge used in the rent-robustness appendix.

Inflation and aggregation bias. Let $I_{q,y,m}^d$ denote the chained Laspeyres index for quintile q at digit depth d . Inflation in 2022 is calculated from annual-average indices,

$$\pi_{q,2022}^d = 100 \left(\frac{\frac{1}{12} \sum_{m=1}^{12} I_{q,2022,m}^d}{\frac{1}{12} \sum_{m=1}^{12} I_{q,2021,m}^d} - 1 \right). \quad (20)$$

The gap is $G^d = \pi_{Q1,2022}^d - \pi_{Q5,2022}^d$ and the microdata aggregation bias is $B^{d,d'} = G^d - G^{d'}$ for $d' > d$. Thus, negative bias indicates a larger gap at finer digit depth.

The micro–macro comparison also reflects differences in source, matched products and income-group construction; it should not be interpreted as aggregation bias. The earlier validation of published HBS expenditure shares uses native digit-3 variables, whereas the inflation calculations aggregate the finest available expenditure variables within each country. In Belgium and the Netherlands, these constructions can differ, notably for insurance expenditure.

Table A.1: Aggregation bias excluding rents: 2022 annual-average inflation

Country	Inflation gaps (pp)					Aggregation bias (pp)			Coverage (%)				
	Macro D3	D2	D3	D4	D5	D2–D3	D2–D4	D2–D5	Macro D3	D2	D3	D4	D5
DE	1.4	2.0	1.5	–	–	0.6	–	–	99.6	100.0	100.0	–	–
FR	0.2	0.3	0.3	–	–	0.0	–	–	99.6	100.0	99.9	–	–
ES	1.0	0.8	1.0	–	–	-0.2	–	–	99.6	100.0	99.9	–	–
AT	0.1	0.5	0.0	0.0	0.0	0.5	0.6	0.6	99.5	100.0	100.0	99.6	97.1
BE	2.3	2.5	2.3	2.4	2.3	0.1	0.0	0.1	99.3	100.0	100.0	99.3	96.9
CY	0.8	0.7	0.8	0.9	1.1	-0.1	-0.2	-0.3	99.8	100.0	100.0	99.8	98.7
HR	0.4	0.1	0.3	0.2	0.3	-0.2	0.0	-0.1	99.7	100.0	100.0	99.9	99.6
EE	4.4	3.0	4.9	4.1	3.8	-1.9	-1.2	-0.8	99.3	100.0	99.9	99.5	97.2
FI	-0.9	-0.6	-0.8	-0.6	-0.8	0.2	0.0	0.2	99.3	100.0	99.7	98.2	89.2
EL	1.4	1.4	0.9	0.8	0.8	0.5	0.7	0.6	99.2	100.0	99.5	99.3	99.0
IE	3.0	2.0	2.9	2.3	2.1	-0.8	-0.3	-0.1	99.7	100.0	100.0	99.9	97.7
LV	4.8	2.9	3.7	3.3	2.0	-0.8	-0.3	1.0	99.4	100.0	97.9	87.3	82.6
LT	3.4	2.9	3.7	4.9	4.9	-0.7	-2.0	-2.0	99.2	100.0	100.0	99.8	98.6
LU	0.2	-0.4	0.2	1.4	1.0	-0.6	-1.8	-1.4	99.7	100.0	100.0	99.9	96.3
MT	0.3	0.6	0.4	0.3	-0.5	0.3	0.4	1.1	99.6	100.0	100.0	98.8	95.0
NL	2.8	2.0	3.0	2.5	1.9	-1.1	-0.6	0.1	99.3	100.0	99.9	99.7	87.8
SK	1.2	0.5	0.1	0.1	0.1	0.4	0.4	0.4	100.0	100.0	100.0	99.9	92.4
SI	0.5	0.5	0.6	0.6	0.7	-0.1	-0.1	-0.2	99.7	100.0	99.8	99.5	99.0

Notes: Inflation gaps and differences are in percentage points; coverage is in percent. Gaps are Q1 minus Q5; D2–D3, D2–D4 and D2–D5 compare micro estimates. Macro D3 uses national-quintile Eurostat HBS aggregates; micro uses survey-weighted total-income quintiles. All columns exclude rents and use RAS on quintile baskets. Construction and coverage are described above. A dash denotes unavailable detail; differences can reflect changes in matched products as well as aggregation.

B Country Exposure to Extra-EU Import-Price Shocks

For shock family k , let h_k denote the observed change in the annual-average extra-EU27 import unit-value index. The energy shock is 83.9 percent and the food shock is 21.0 percent between 2021 and 2022. These are common external-price innovations: country heterogeneity enters through import exposure rather than through different values of h_k .

For destination country d and industry j , the raw import-price shock becomes the imported-input cost impulse

$$b_{dj}^k = h_k \frac{\sum_{o \notin EU27} \sum_{i \in \mathcal{S}_k} Z_{oi,dj}}{x_{dj}}, \quad (21)$$

where $\mathcal{S}_k$ is the set of shocked products, $Z_{oi,dj}$ is the intermediate-input flow from origin o and product i to industry j in country d , and x_{dj} is that industry's gross output. Thus, an identical raw price shock produces different cost impulses across countries and industries because their extra-EU imported-input shares differ.

The output-weighted country impulse reported in Table B.1 is

$$\bar{b}_d^k = h_k \frac{\sum_j \sum_{o \notin EU27} \sum_{i \in \mathcal{S}_k} Z_{oi,dj}}{\sum_j x_{dj}}. \quad (22)$$

The euro-area row applies this ratio to EA20 sums; it is not a simple average of the country rows.

Table B.1: Import exposure and imported-input cost impulses by euro-area country, 2022

Country	Energy products		Food products	
	Raw price shock: 83.9%		Raw price shock: 21.0%	
	M/x	b	M/x	b
Austria	0.79	0.66	0.11	0.02
Belgium	2.36	1.98	0.24	0.05
Croatia	2.62	2.20	0.24	0.05
Cyprus	0.62	0.52	0.14	0.03
Estonia	0.87	0.73	0.29	0.06
Finland	2.27	1.91	0.13	0.03
France	1.52	1.27	0.12	0.02
Germany	1.20	1.01	0.10	0.02
Greece	6.51	5.46	0.22	0.05
Ireland	0.78	0.65	0.30	0.06
Italy	2.00	1.68	0.29	0.06
Latvia	0.73	0.61	0.23	0.05
Lithuania	6.07	5.09	0.23	0.05
Luxembourg	0.07	0.06	0.05	0.01
Malta	0.81	0.68	0.10	0.02
Netherlands	2.60	2.18	0.62	0.13
Portugal	2.47	2.07	0.42	0.09
Slovakia	2.96	2.48	0.16	0.03
Slovenia	0.96	0.81	0.16	0.03
Spain	2.70	2.27	0.29	0.06
Euro area	1.80	1.51	0.21	0.04

Notes. The raw shocks are the common changes in the 2022 annual-average extra-EU27 import unit-value indices relative to 2021. M/x is the extra-EU imported-input share of gross output, and $b = h(M/x)$ is the corresponding cost impulse; both columns are expressed in percent. Energy covers B and C19; food covers A01 and C10–12. The euro-area entries are ratios of EA20 sums. Sources: Eurostat international trade statistics, FIGARO, authors' calculations.

Table B.2 repeats the simulation for each annual energy import-price change from 2021 to 2023 and reports the resulting effects by country. To isolate changes in the external shocks, the simulation holds the 2022 FIGARO network and all 2022 expenditure and country weights fixed across the three columns.

Table B.2: Simulated country effects of the observed extra-EU energy import-price shock, 2021–2023

Country	Effect on average inflation			Effect on inflation inequality		
	2021	2022	2023	2021	2022	2023
Germany	4.0	5.6	-1.8	0.9	1.2	-0.4
France	5.1	7.1	-2.3	1.3	1.8	-0.6
Italy	5.8	8.1	-2.6	0.5	0.7	-0.2
Spain	4.3	5.9	-1.9	0.7	0.9	-0.3
Netherlands	5.6	7.9	-2.5	0.9	1.2	-0.4
Belgium	3.8	5.2	-1.7	0.5	0.7	-0.2
Austria	2.6	3.5	-1.1	0.2	0.2	-0.1
Portugal	4.9	6.8	-2.2	0.7	1.0	-0.3
Greece	6.1	8.6	-2.7	0.5	0.7	-0.2
Finland	3.1	4.4	-1.4	-0.7	-1.0	0.3
Ireland	3.8	5.3	-1.7	2.5	3.5	-1.1
Slovakia	4.6	6.5	-2.1	1.4	2.0	-0.6
Croatia	6.0	8.4	-2.7	0.5	0.7	-0.2
Lithuania	6.2	8.6	-2.8	0.2	0.3	-0.1
Slovenia	4.0	5.5	-1.8	0.1	0.1	-0.0
Luxembourg	1.6	2.2	-0.7	0.3	0.5	-0.2
Latvia	2.7	3.7	-1.2	0.7	1.0	-0.3
Estonia	3.3	4.6	-1.5	0.3	0.4	-0.1
Cyprus	4.1	5.7	-1.8	1.6	2.2	-0.7
Malta	2.4	3.3	-1.0	-0.0	-0.0	0.0
Euro area	4.6	6.5	-2.1	0.8	1.1	-0.4

Notes. Entries are percentage-point effects of the energy import-price shock. Each column uses the change in the annual-average extra-EU27 energy import unit-value index relative to the previous year. The 2022 FIGARO network, product weights, household weights, and country weights are held fixed across years. Countries are ordered by their 2022 HICP country weight. Sources: Eurostat international trade statistics, HBS, FIGARO, authors' calculations.

C Additional Figures

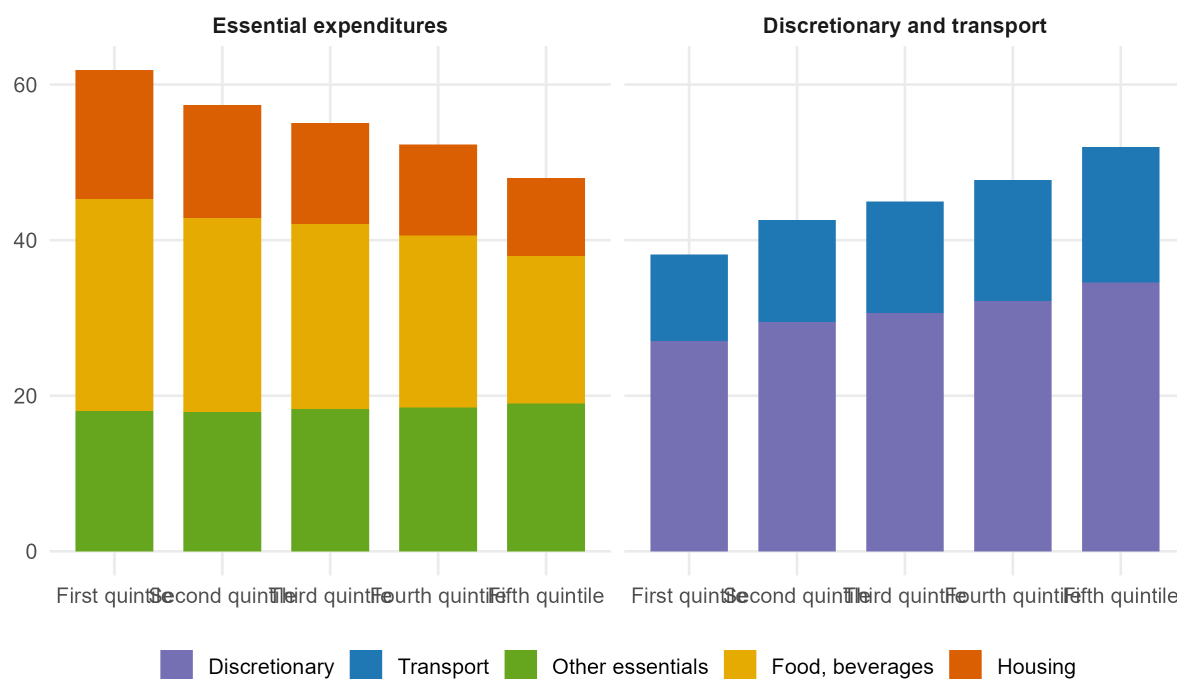
Figure C.1: Main COICOP expenditure weights by income quintile, Spain (in % of total expenditure)



Notes: The figure reports the share of each main COICOP category in total household expenditure for the first and fifth equivalised-income quintiles. Quintiles are computed separately by year using household weights.

Sources: INE Encuesta de Presupuestos Familiares public-use microdata; authors' calculations.

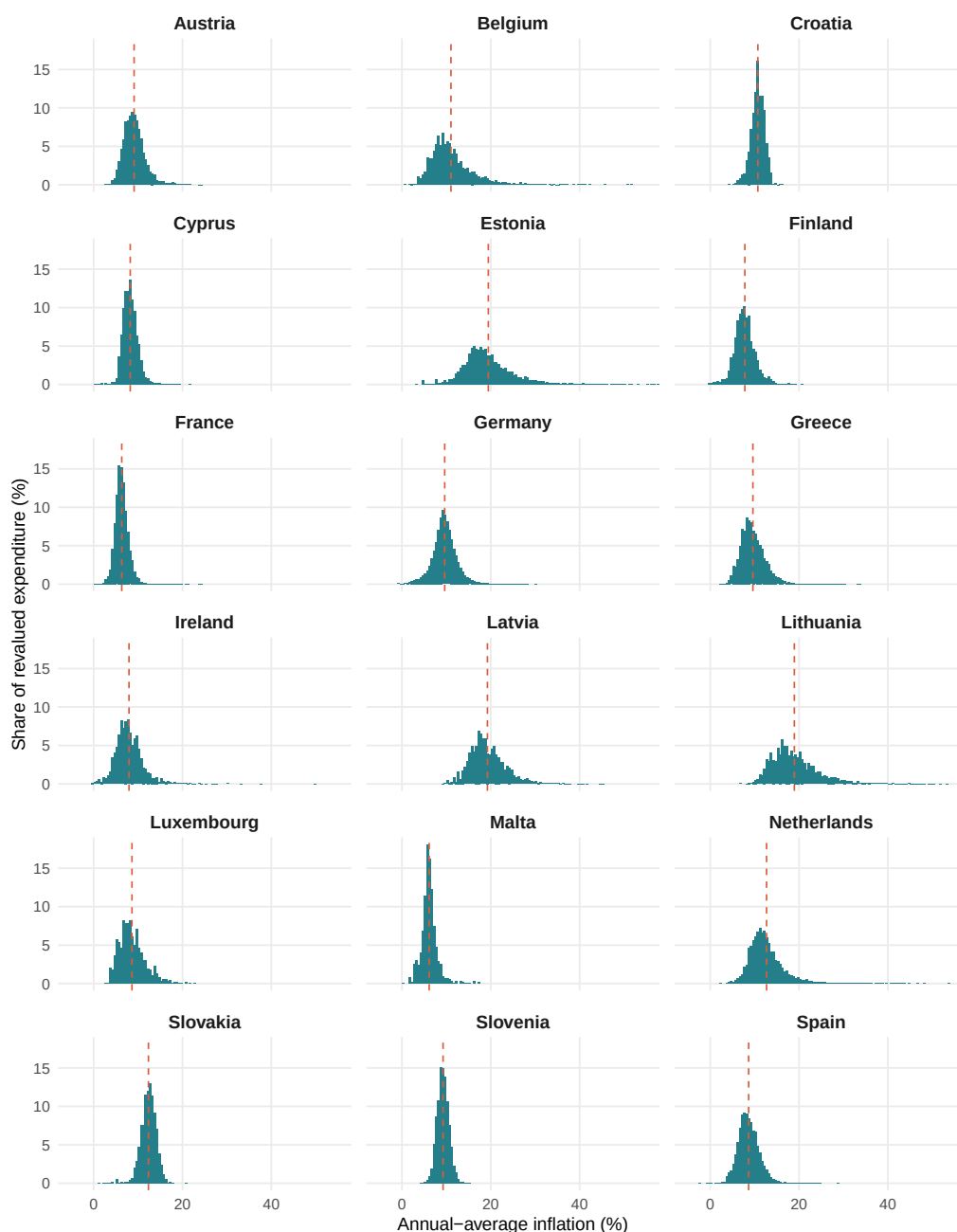
Figure C.2: Essential and discretionary expenditures by income quintile, euro area (in % of total expenditure)



Notes: The figure partitions COICOP expenditure shares into essential expenditures and discretionary/transport expenditures. The euro-area aggregate is computed as the unweighted average of available country-level HBS expenditure shares.

Sources: Eurostat HBS 2015; authors' calculations.

Figure C.3: National distributions of household inflation in 2022: RAS, COICOP digit 3



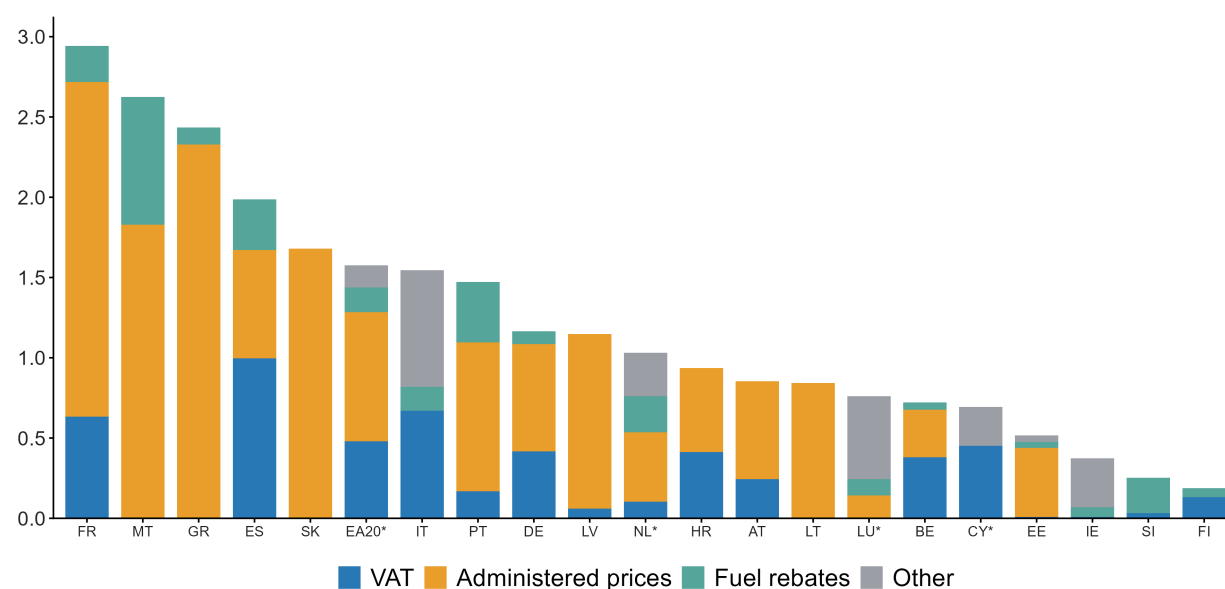
Notes: Annual-average inflation; 2020 HBS baskets, excluding rents (041 and 042). Weights equal survey weights times retained expenditure times the 2021 annual-average household index. Orange lines mark the corresponding weighted means. Bins are 0.5 percentage points wide; the displayed range is -5 to 55 percent, without renormalization. The expenditure share outside this range is below 0.1 percent in every country. The sample comprises 152,613 households in 18 countries, excluding Italy and Portugal.

Sources: HBS microdata, Eurostat HICP; authors' calculations.

D Counterfactuals

This appendix reports country-level observed and counterfactual headline inflation series. Each figure aggregates all simulated energy and food price-policy layers available for the country, using the HICP item weights in the country CPI basket.

Figure D.1: Fiscal Costs of Energy Measures (Percent of 2022 GDP)



Notes: Bars sum documented costs of the measures active in the counterfactual during 2021–2023, counting each shared budget envelope once. Fuel price reductions comprise all road-fuel (CP0722) relief, including excise-duty reductions, regulated-price support, and direct subsidies; the VAT and non-fuel excise category therefore excludes road-fuel measures. Other comprises remaining instruments and shared envelopes with no identified instrument split. An asterisk indicates an unidentified instrument split. All active measures have a documented cost, but budget reference periods vary across schemes and do not uniformly cover 2021–2023; costs are not restricted to amounts received by households. EA20 divides summed costs by summed nominal 2022 GDP.

Sources: Bruegel, national official sources, Eurostat; authors' calculations using the fiscal-cost registry underlying the policy tables.

Table D.1: Price measures included in the national counterfactual simulations

Country	Simplified policy coverage	Policy years	Effect identification	Policy count
Austria	Products: electricity, gas. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.	2022–2024	0 source-determined; 4 model-	4 entries; 3 distinct policies imputed.
Belgium	Products: electricity, gas, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.	2022–2023	2 source-determined; 4 model-	6 entries; 3 distinct policies imputed.
Croatia	Products: food, electricity, gas, heating. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.	2022–2023	7 source-determined; 2 model-	9 entries; 4 distinct policies imputed.
Cyprus	Products: electricity, heating, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.	2021–2023	1 source-determined; 4 model-	5 entries; 3 distinct policies imputed.
Estonia	Products: electricity, gas, heating, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.	2021–2024	0 source-determined; 8 model-	8 entries; 8 distinct policies imputed.
Finland	Products: electricity, motor fuels, passenger transport. Instruments: tax and charge relief.	2022–2023	5 source-determined; 1 model-	6 entries; 3 distinct policies imputed.
France	Products: electricity, gas, motor fuels. Instruments: price caps or regulated tariffs, subsidies or rebates.	2021–2024	2 source-determined; 1 model-	3 entries; 3 distinct policies imputed.
Germany	Products: electricity, gas, heating, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.	2022–2024	1 source-determined; 6 model-	7 entries; 6 distinct policies imputed.
Greece	Products: electricity, gas, motor fuels. Instruments: subsidies or rebates.	2021–2023	0 source-determined; 3 model-	3 entries; 3 distinct policies imputed.
Ireland	Products: electricity, gas, motor fuels. Instruments: tax and charge relief, subsidies or rebates.	2022–2023	2 source-determined; 3 model-	5 entries; 5 distinct policies imputed.

Continued on next page

Country	Simplified policy coverage	Policy years	Effect identification	Policy count
Italy	Products: electricity, gas, motor fuels. Instruments: tax and charge relief, 2021–2023 subsidies or rebates.		2 source-determined; 4 model-	6 entries; 4 distinct policies imputed.
Latvia	Products: electricity, gas, heating. Instruments: price caps or regulated tariffs, 2022–2023 tax and charge relief, subsidies or rebates.		0 source-determined; 3 model-	3 entries; 3 distinct policies imputed.
Lithuania	Products: electricity, gas. Instruments: subsidies or rebates.	2022–2023	0 source-determined; 2 model-	2 entries; 1 distinct policy imputed.
Luxembourg	Products: electricity, gas, heating, motor fuels, other consumption. Instruments: 2022–2024 price caps or regulated tariffs, tax and charge relief, subsidies or rebates.		67 source-determined; 5 model-	72 entries; 6 distinct policies imputed.
Malta	Products: electricity, motor fuels. Instruments: price caps or regulated tariffs, 2022–2023 subsidies or rebates.		0 source-determined; 2 model-	2 entries; 2 distinct policies imputed.
Netherlands	Products: electricity, gas, heating, motor fuels. Instruments: price caps or 2022–2023 regulated tariffs, tax and charge relief.		7 source-determined; 4 model-	11 entries; 6 distinct policies imputed.
Portugal	Products: food, electricity, gas, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief.	2021–2026	1 source-determined; 12 model-	13 entries; 6 distinct policies imputed.
Slovakia	Products: electricity, gas, heating. Instruments: price caps or regulated tariffs, 2023 tax and charge relief.		0 source-determined; 3 model-	3 entries; 1 distinct policy imputed.
Slovenia	Products: electricity, gas, heating, motor fuels. Instruments: price caps or 2021–2023 regulated tariffs, tax and charge relief.		0 source-determined; 9 model-	9 entries; 7 distinct policies imputed.
Spain	Products: food, electricity, gas, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.	2021–2026	5 source-determined; 10 model-	15 entries; 9 distinct policies imputed.
Euro area	Sum of the 20 country rows.	2021–2026	102 source-determined; 90 model-	192 entries; 86 distinct policies imputed.

Notes: Products and instruments are grouped for readability; the underlying workbook retains the measure-level descriptions, dates, COICOP codes, and the measure-specific evidence basis. Source-determined effects are quantified directly from statutory tax rates, regulated tariffs, or official administrative price or bill series. Model-imputed effects require an aggregate-equivalent wedge, representative bill, calibrated pass-through, or model-based proxy. An official policy announcement alone does not make an effect source-determined when its price effect must be estimated. Policy count reports executed country-product entries and distinct policy devices. The distinct count consolidates successive phases or extensions and a single legal measure represented by several product trajectories; for example, Luxembourg's product-specific VAT trajectories count as one policy. The Euro-area row is the sum of the 20 country rows. Sources: Bruegel Dataset (2021), 'National policies to shield consumers from rising energy prices,' version of 26 June 2023; authors' calculations.

D.1 Austria

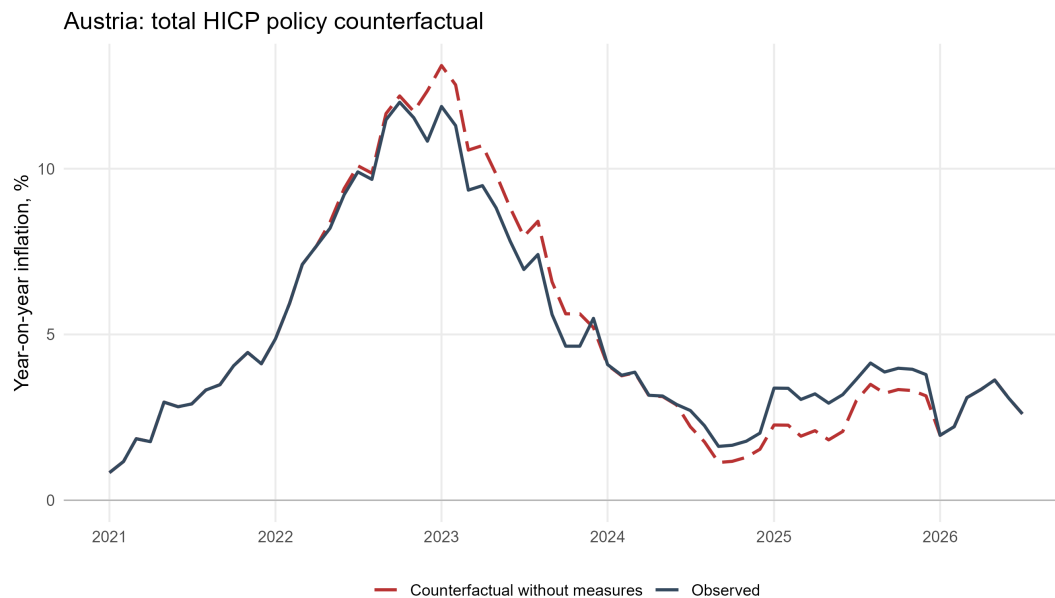


Figure D.2: Austria: observed and counterfactual inflation

D.2 Belgium

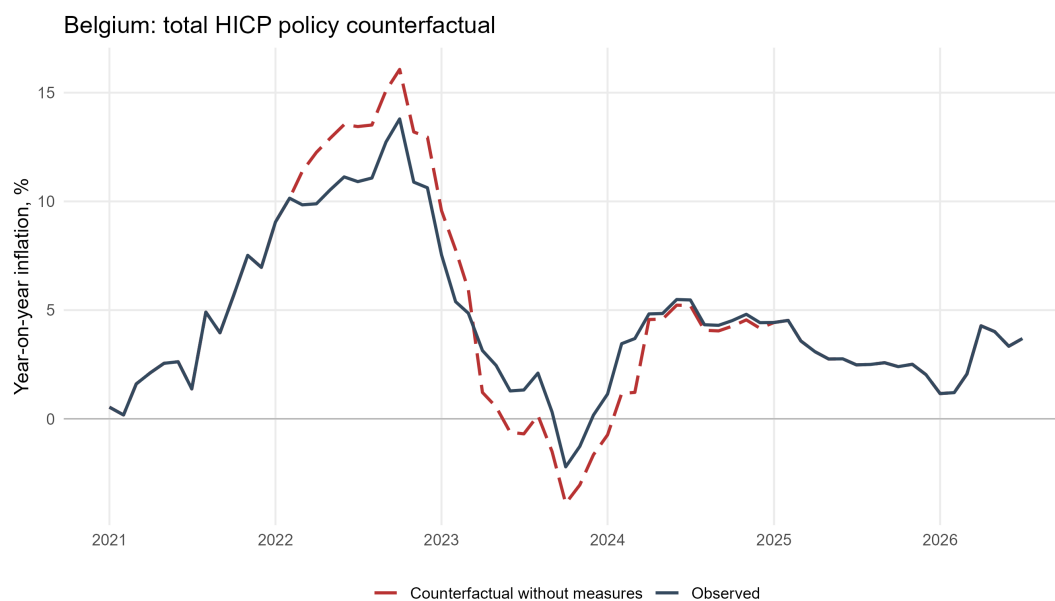


Figure D.3: Belgium: observed and counterfactual inflation

D.3 Cyprus

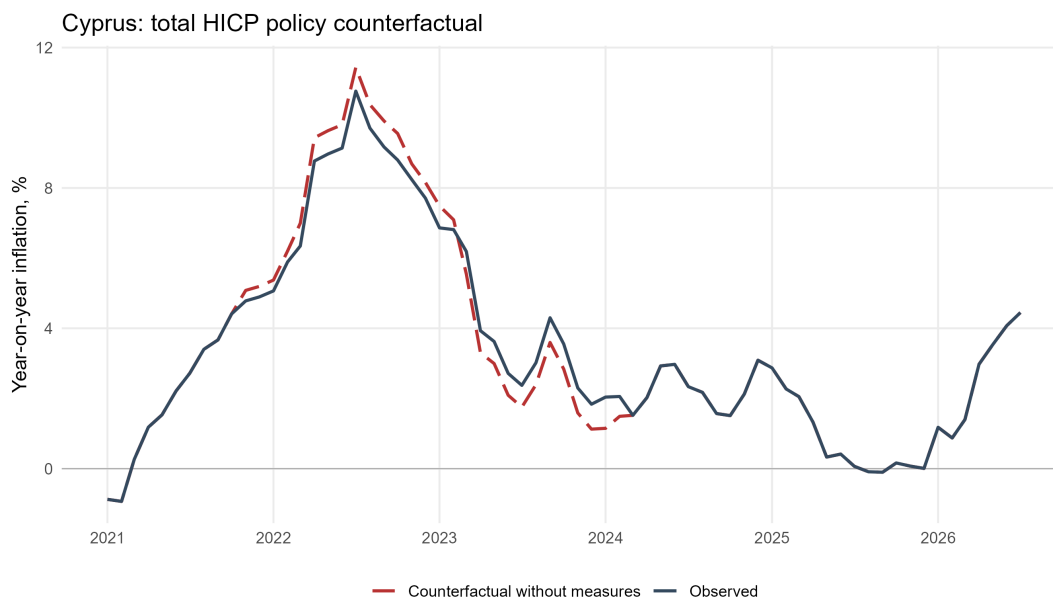


Figure D.4: Cyprus: observed and counterfactual inflation

D.4 Germany

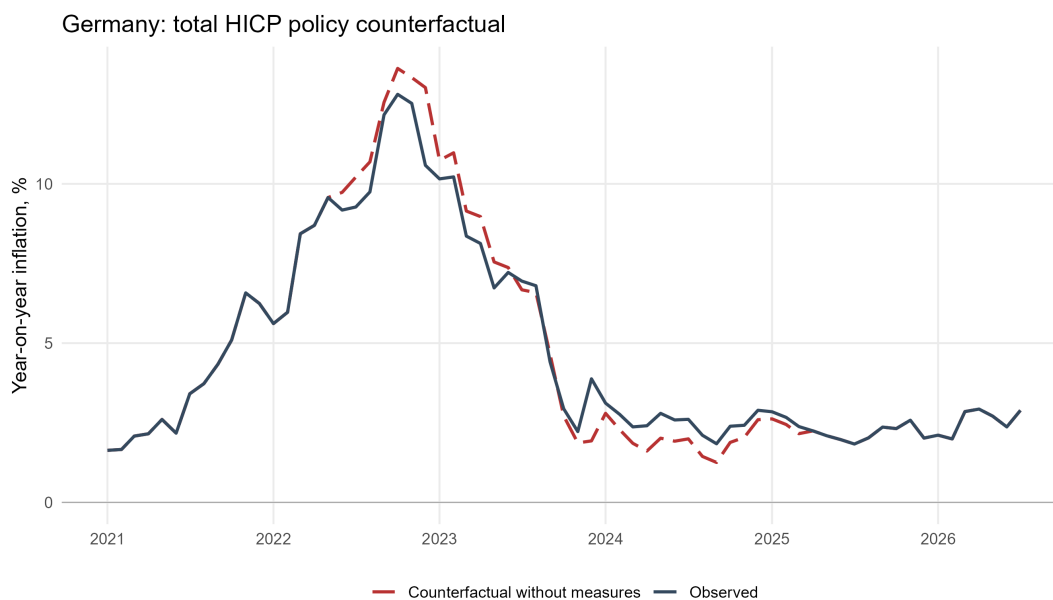


Figure D.5: Germany: observed and counterfactual inflation

D.5 Estonia

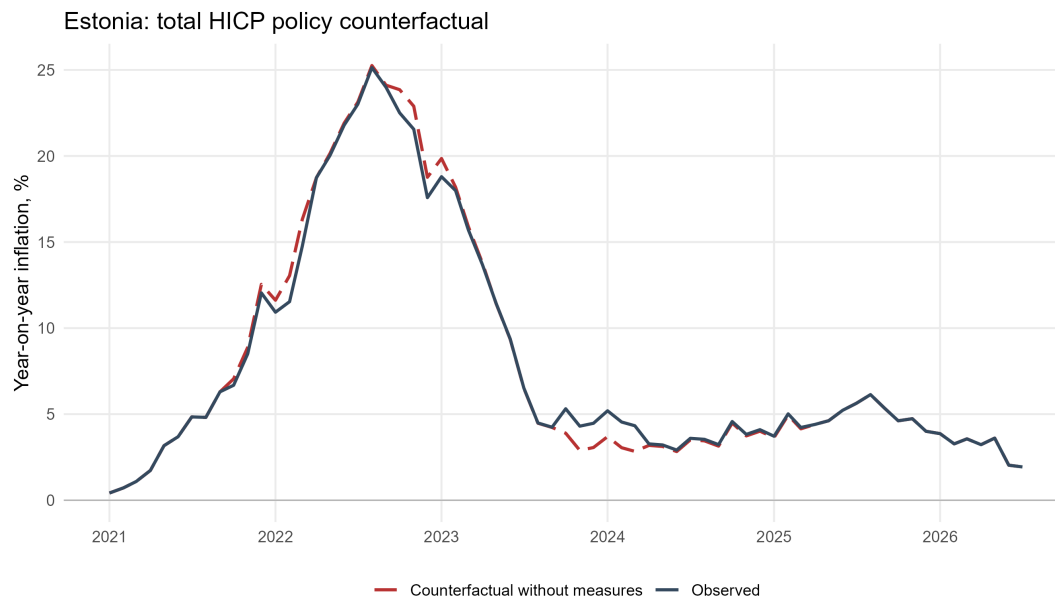


Figure D.6: Estonia: observed and counterfactual inflation

D.6 Greece

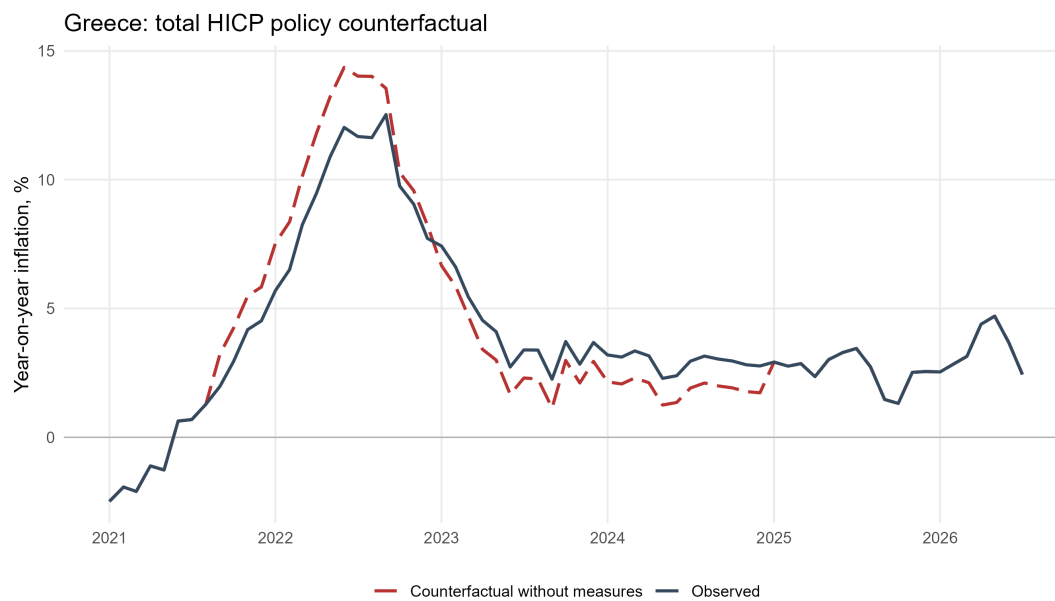


Figure D.7: Greece: observed and counterfactual inflation

D.7 Spain

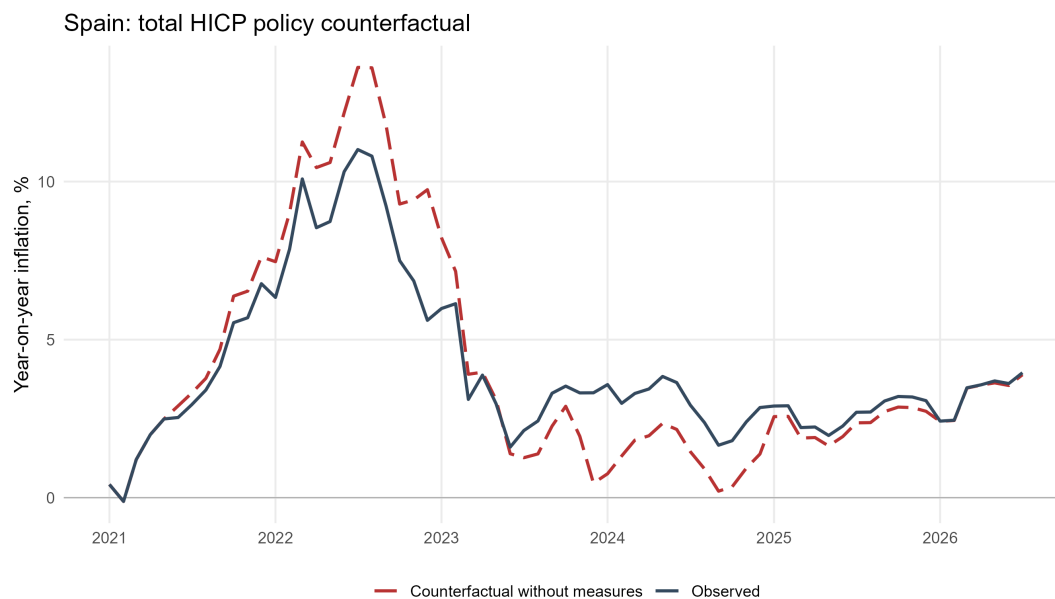


Figure D.8: Spain: observed and counterfactual inflation

D.8 Finland

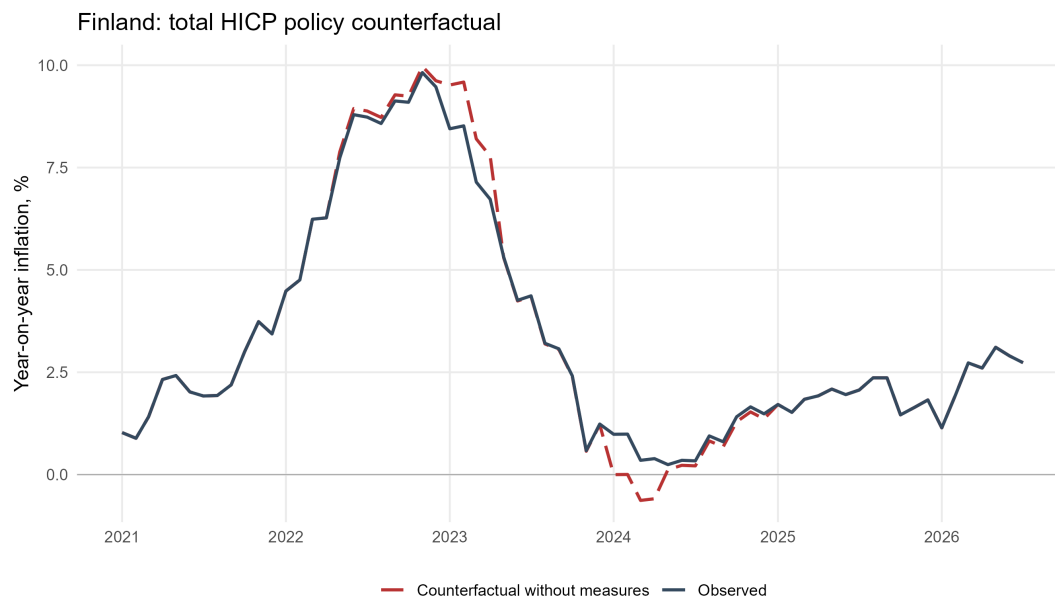


Figure D.9: Finland: observed and counterfactual inflation

D.9 France

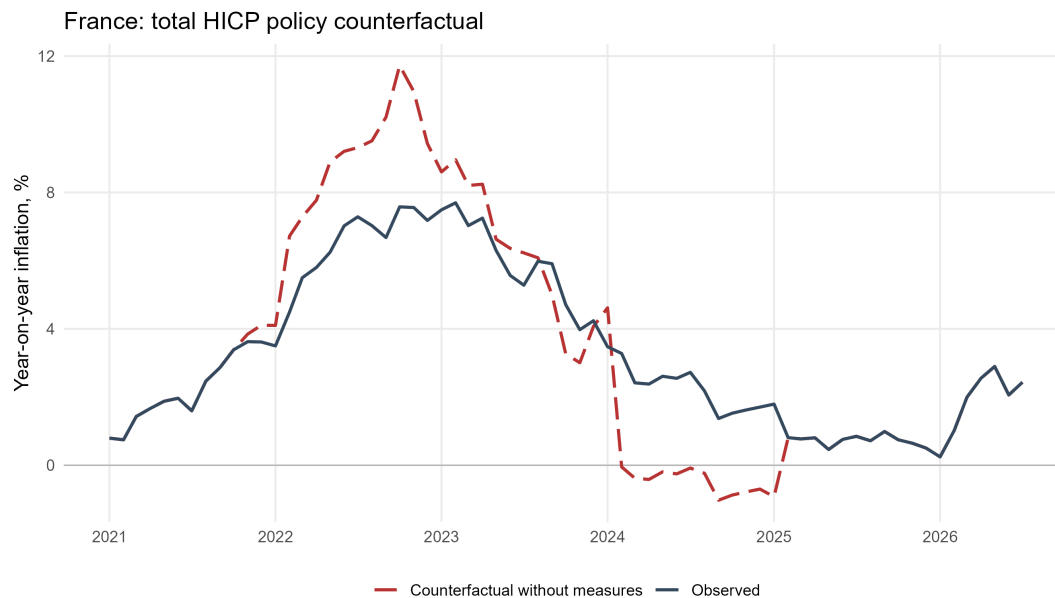


Figure D.10: France: observed and counterfactual inflation

D.10 Croatia

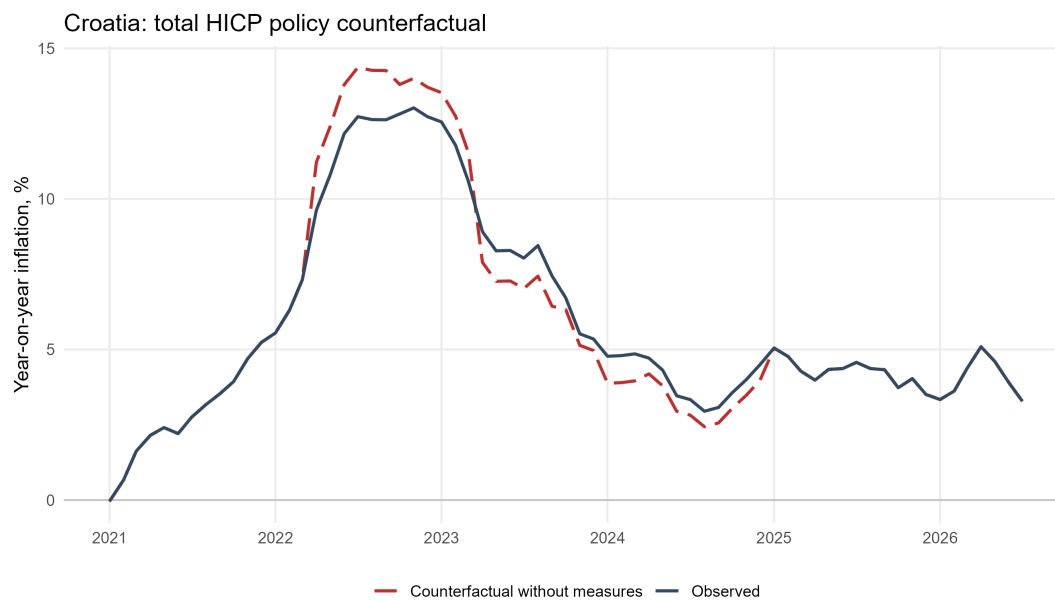


Figure D.11: Croatia: observed and counterfactual inflation

D.11 Ireland

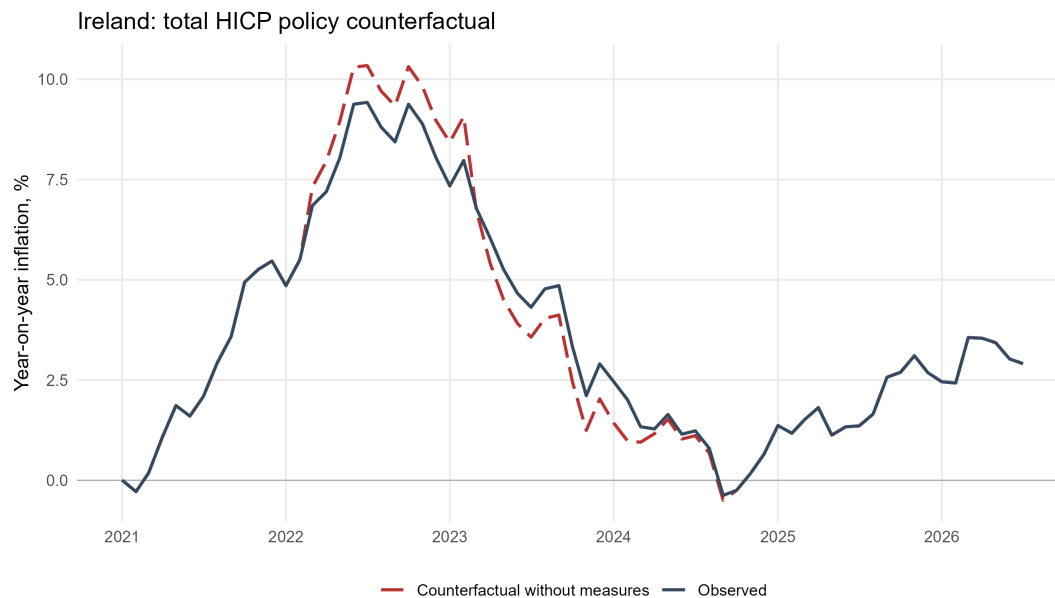


Figure D.12: Ireland: observed and counterfactual inflation

D.12 Italy

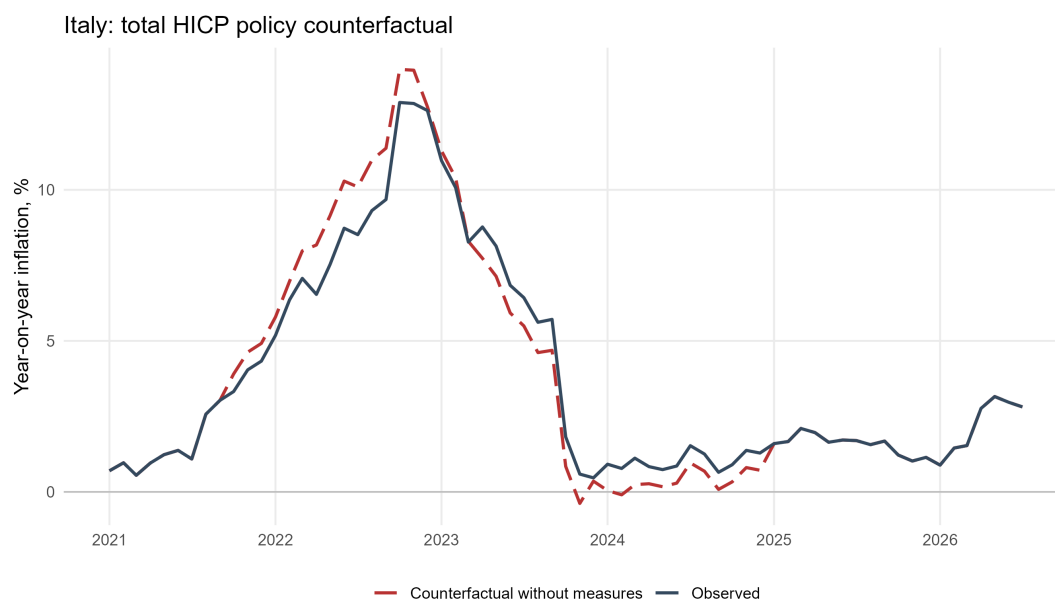


Figure D.13: Italy: observed and counterfactual inflation

D.13 Lithuania

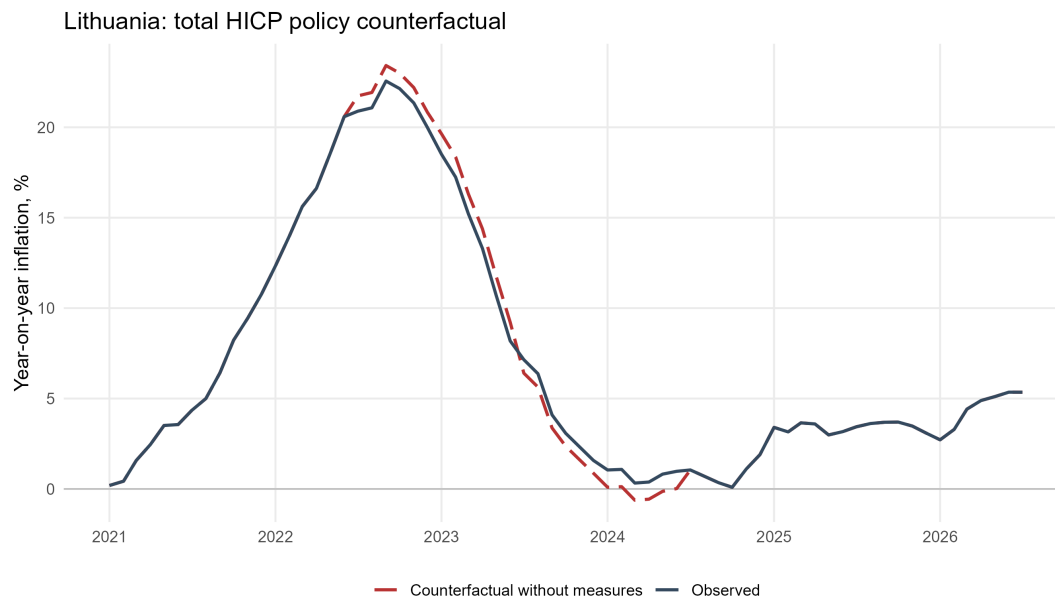


Figure D.14: Lithuania: observed and counterfactual inflation

D.14 Luxembourg

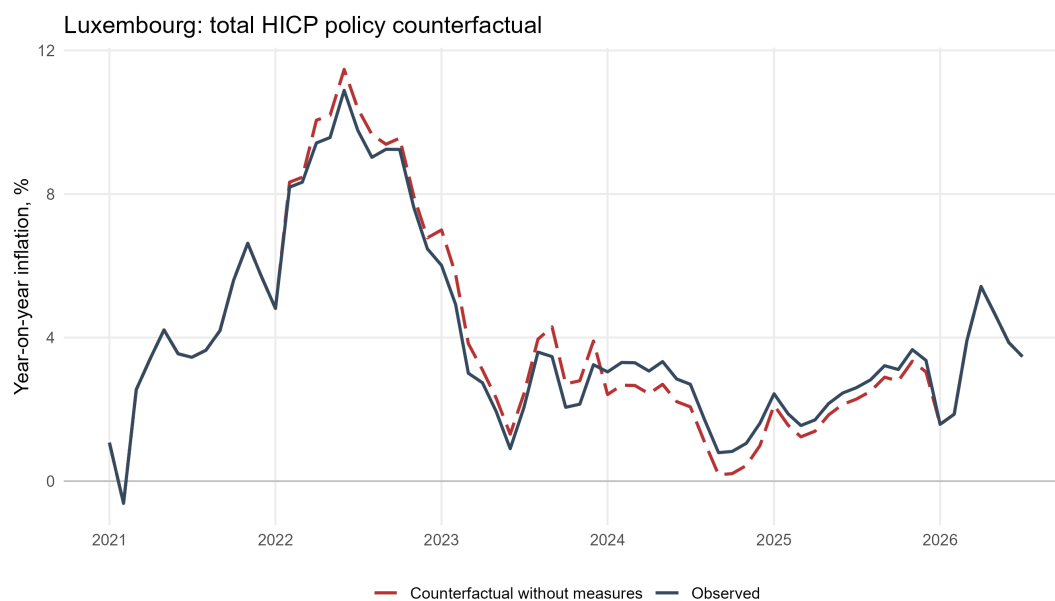


Figure D.15: Luxembourg: observed and counterfactual inflation

D.15 Latvia

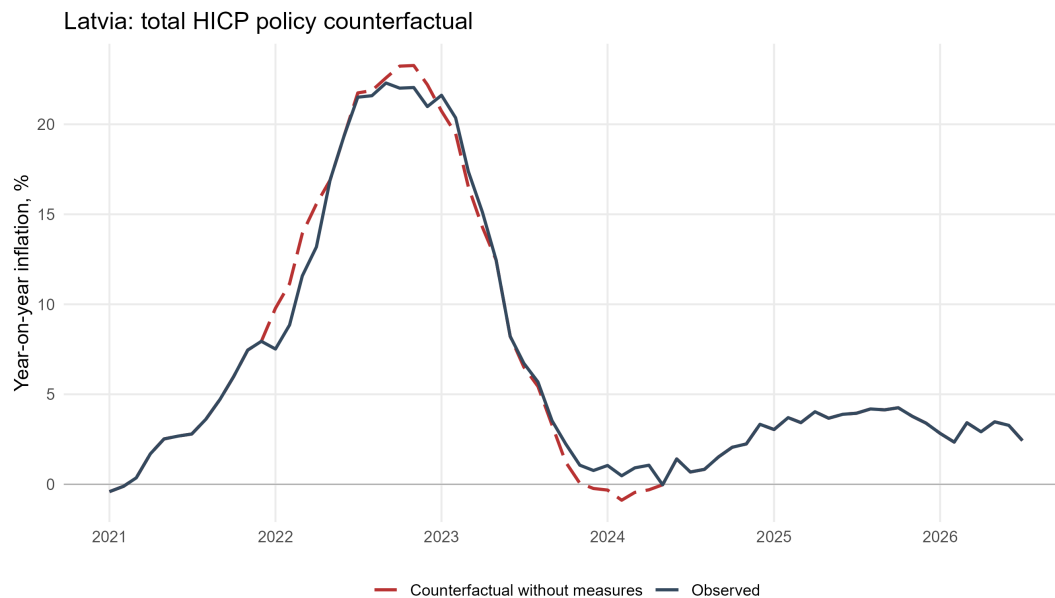


Figure D.16: Latvia: observed and counterfactual inflation

D.16 Malta

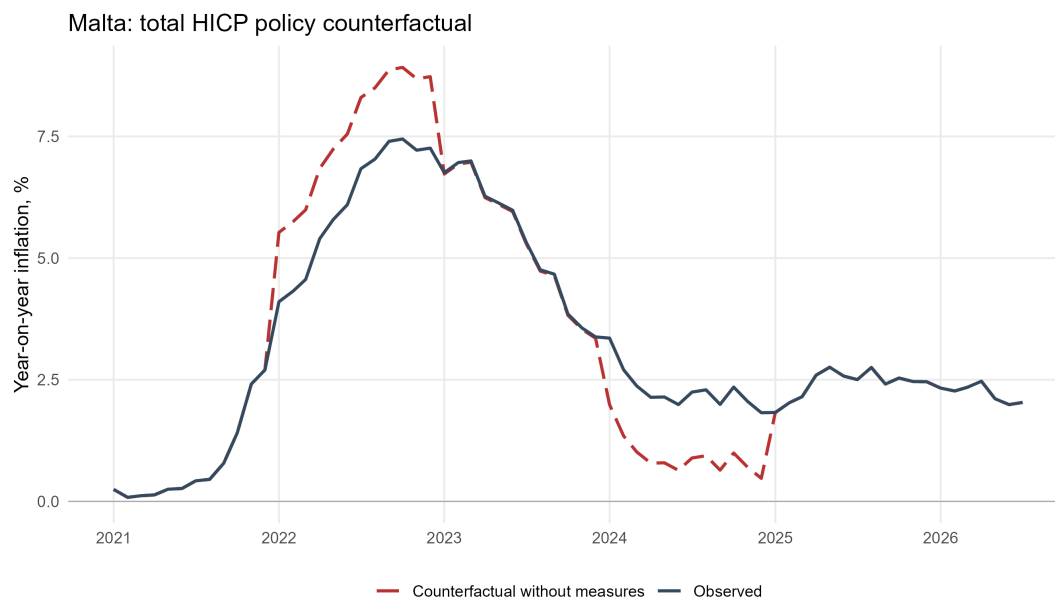


Figure D.17: Malta: observed and counterfactual inflation

D.17 Netherlands

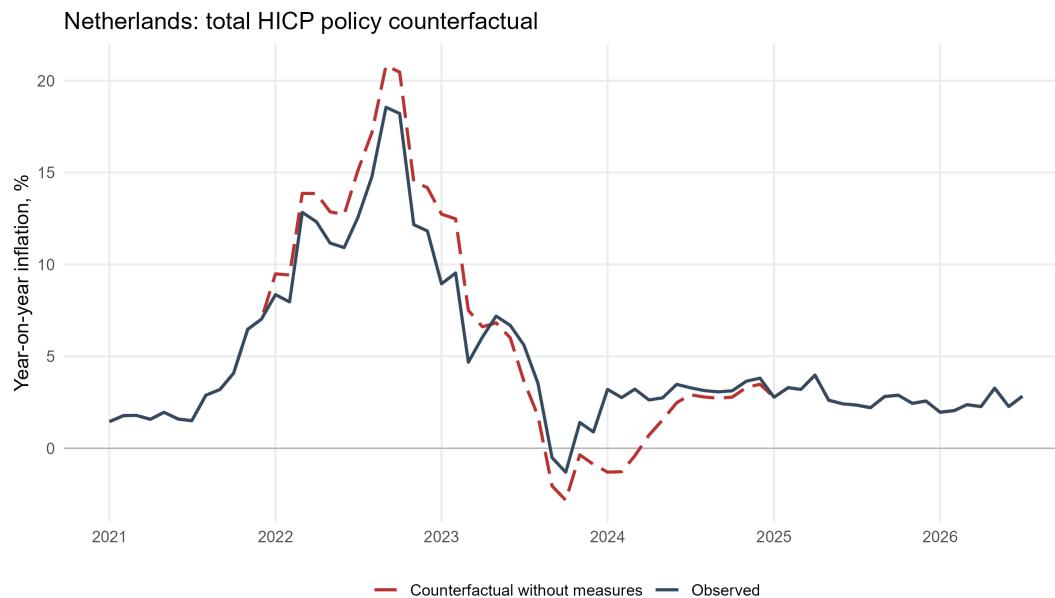


Figure D.18: Netherlands: observed and counterfactual inflation

D.18 Portugal

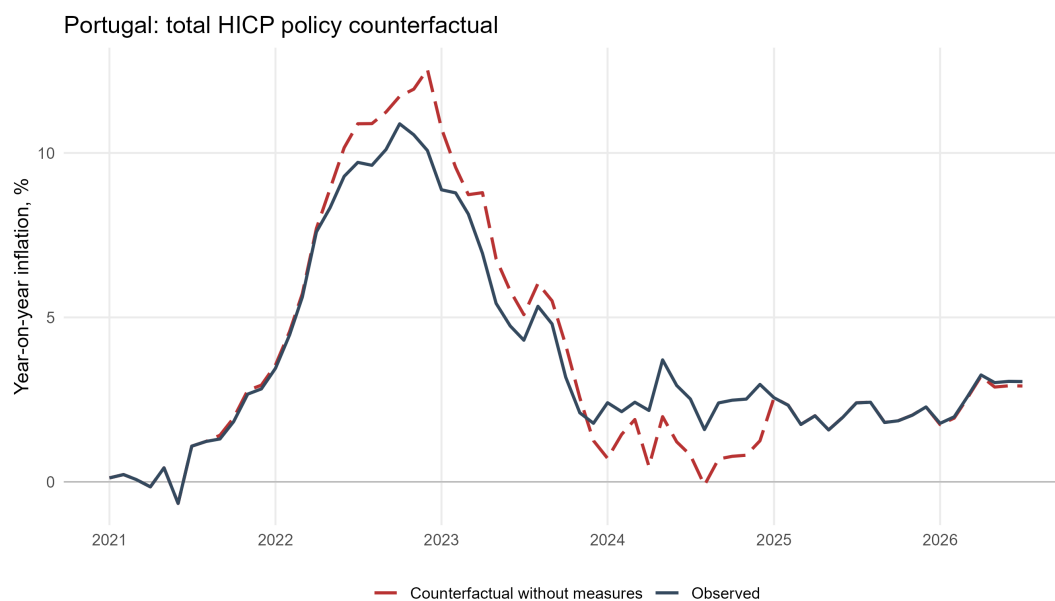


Figure D.19: Portugal: observed and counterfactual inflation

D.19 Slovakia

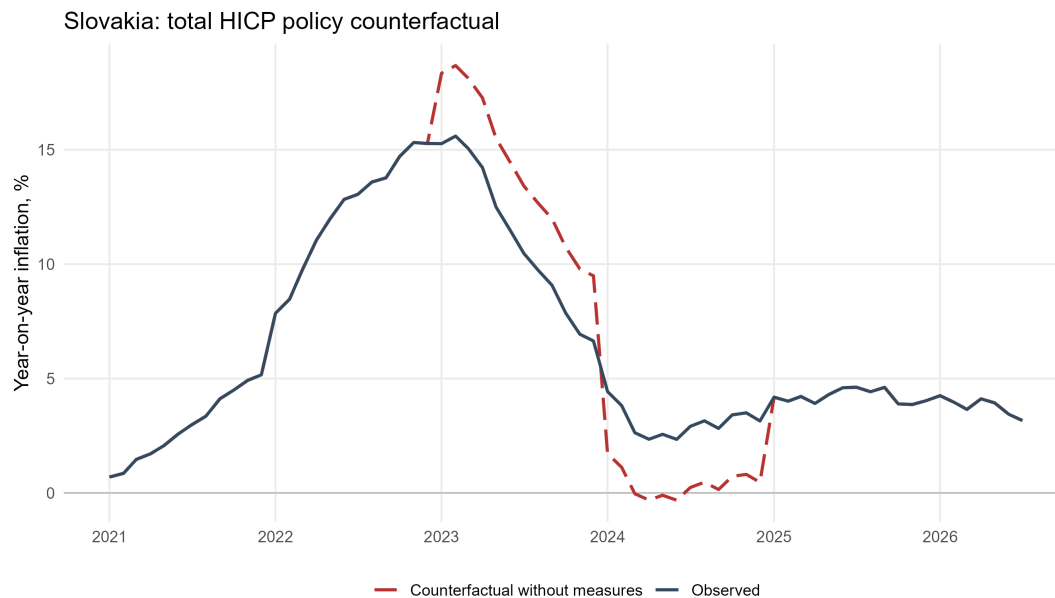


Figure D.20: Slovakia: observed and counterfactual inflation

D.20 Slovenia

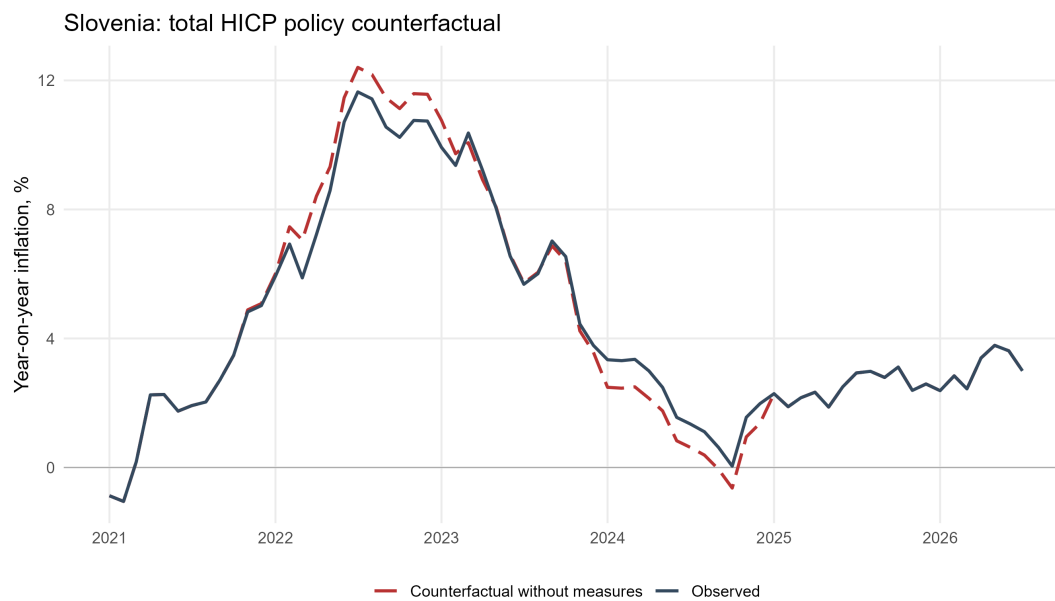


Figure D.21: Slovenia: observed and counterfactual inflation

E The Household Budget Survey in Italy

The most recent complete harmonized Eurostat HBS entry for Italy is 2005. We therefore combine it with the 2015 and 2020 Istat public-use HBS files to construct the household-group baskets.

Age-group baskets are computed directly from the survey-weighted expenditure aggregates in the Istat files, using the age of the household reference person.

For income quintiles, households are ranked by equivalised total consumption using the Carbonaro scale and the survey weights. Because income quintiles are not observed in the target-year files, we transport the relationship between income- and expenditure-quintile baskets observed in 2005. Let E_{jqy} denote expenditure on COICOP group j by expenditure quintile q in year y , and let $s_{jq,2005}^I$ and $s_{jq,2005}^E$ denote the corresponding 2005 shares for income and expenditure quintiles. The estimated income-quintile consumption structure is

$$\hat{s}_{jqy}^I = \frac{E_{jqy} \left(s_{jq,2005}^I / s_{jq,2005}^E \right)}{\sum_k E_{kqy} \left(s_{kq,2005}^I / s_{kq,2005}^E \right)}. \quad (23)$$

Thus, the 2005 income-to-expenditure ratio adjusts each 2015 or 2020 expenditure-quintile component, after which the basket is renormalized to one. A few broad groups are combined to reconcile the two 2005 classifications. The resulting income-quintile structures are estimated rather than directly observed. The procedure addresses the same absence of jointly observed income and consumption discussed by [Donatiello et al. \(2022\)](#).

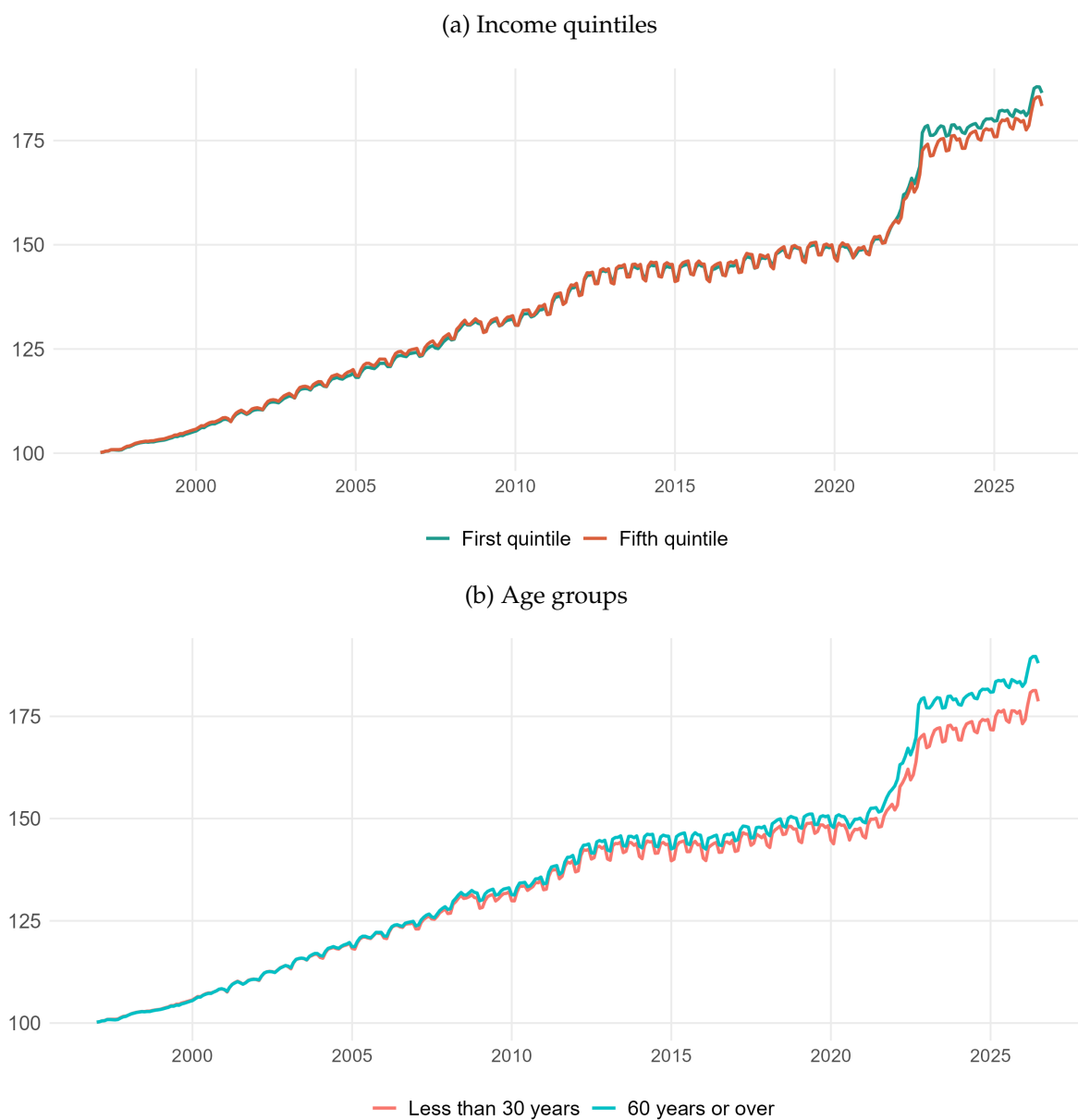
Table [E.1](#) reports the resulting shares of total consumption. Age shares are direct survey estimates; income-quintile shares are bridge estimates. The first and fifth income quintiles account for about 8 and 39 percent of Italian consumption, respectively.

Table E.1: Expenditures by groups (as a share of total expenditures)

Dimension	Group	2015	2020
Age	Less than 30 years	2.0	2.2
Age	From 30 to 44 years	19.7	17.9
Age	From 45 to 59 years	34.8	34.2
Age	60 years or over	43.5	45.7
Income	First quintile	7.8	8.1
Income	Second quintile	12.7	12.8
Income	Third quintile	17.2	17.0
Income	Fourth quintile	23.3	22.8
Income	Fifth quintile	39.0	39.3

Notes. Entries are shares of annual consumption. Age groups follow Eurostat bands; income rows use the Istat–Eurostat quintile bridge. Sources: Istat HBS public-use files, Eurostat HBS, authors’ calculations.

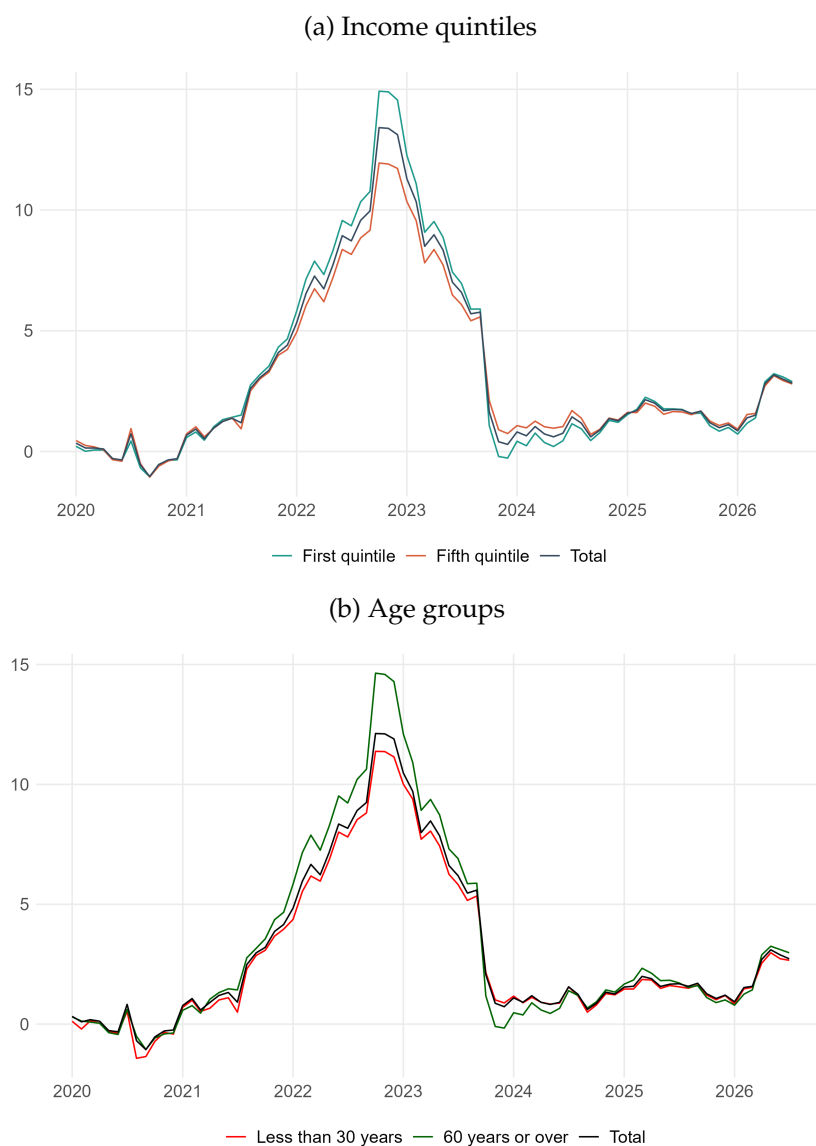
Figure E.1: Price indices by household group, Italy, COICOP digit 3 (index, 1996 = 100)



Notes: The figure reports chained price indices using RAS weighting. Income groups compare the first and fifth quintiles; age groups compare households whose reference person is under 30 with those aged 60 or over.

Sources: Eurostat HICP; Italian HBS inputs described in Appendix E; authors' calculations.

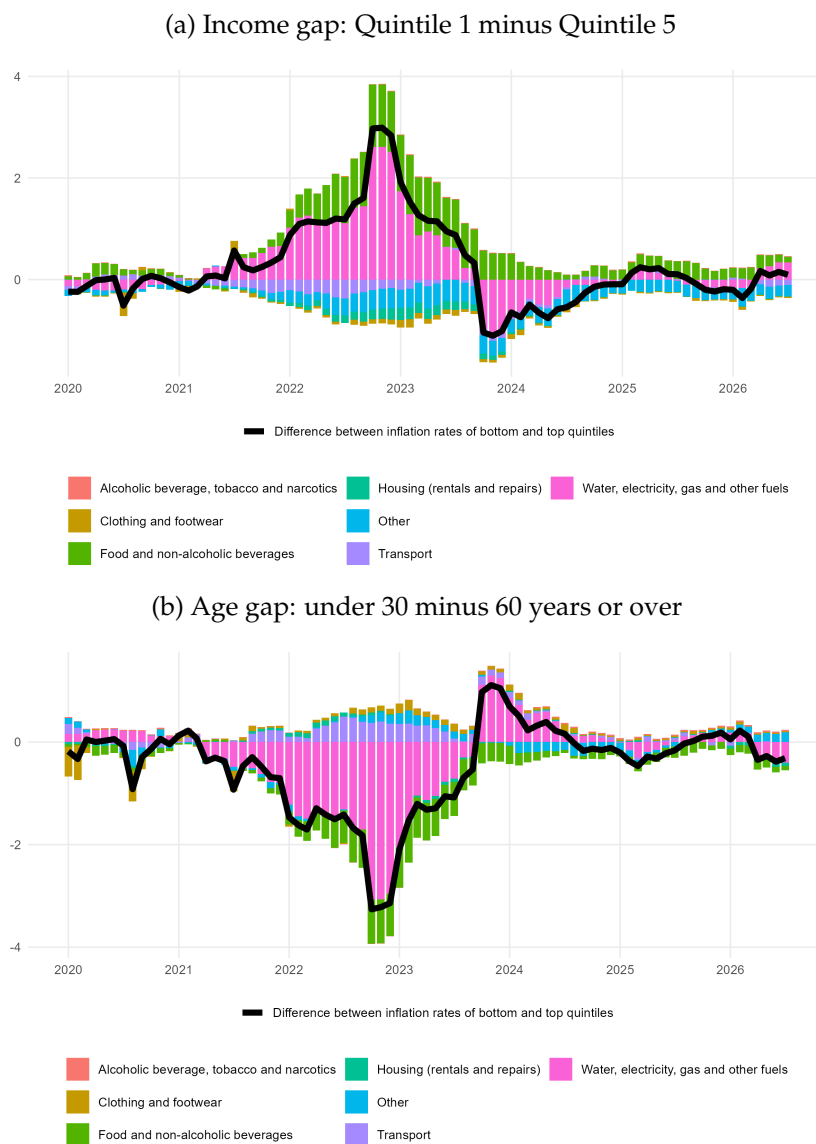
Figure E.2: Inflation by household group, Italy, COICOP digit 3 (in %)



Notes: The figure reports monthly year-on-year inflation rates using RAS weighting for income and age groups.

Sources: Eurostat HICP; Italian HBS inputs described in Appendix E; authors' calculations.

Figure E.3: Contributions to inflation gaps, Italy, COICOP digit 3 (in pp)



Notes: Bars decompose the monthly inflation gap into broad consumption components; the black line reports the total gap. Positive values indicate higher inflation for the first group named in each panel. In panel (b), the black line is the under-30 minus 60-or-over age gap; the generic quintile wording inside the image does not apply to this panel.

Sources: Eurostat HICP; Italian HBS inputs described in Appendix E; authors' calculations.

F RAS Weighting

We use the RAS (Row and Column Adjustment Scaling) algorithm. RAS is a bi-proportional matrix scaling method that alternately rescales rows and columns until the required marginal totals are matched (Bacharach, 1965). In our setting, we combine official HICP item weights with the relative HBS expenditure profile of each household group.

The baseline removes actual and imputed rents (HBS 041 and 042 and corresponding HICP rent components) before constructing matched product baskets. Parent housing aggregates are recalculated without these components. RAS matches group expenditure margins and annual HICP product margins normalized over the remaining matched non-rent products. The resulting indices therefore measure inflation excluding rents; agreement with the all-items HICP is not an identity implied by this calibration.

Rent treatment can affect both the magnitude and the sign of income gaps because housing tenure and rent shares vary across groups (Geerolf, 2023). Table F.1 retains three alternative conventions as a separate sensitivity comparison: all rents maps HBS 041+042 to HICP 041; effective rents retains HBS 041 only; excluding rents removes 041 and 042 and renormalizes the remaining basket. This reference exercise uses relative-expenditure weights. It is distinct from the RAS-based baseline and the RAS micro–macro comparisons in Appendix A; its excluding-rents column need not reproduce their estimates.

We distinguish three objects: the HICP product shares, the aggregate product-by-group expenditure mass calibrated by RAS, and the resulting within-group baskets. First, let $s_{g,y}$ denote the share of group g in total consumption expenditure used for HICP weight year y :

$$s_{g,y} = \frac{E_{g,y'}^{\text{HBS}}}{E_{\text{all},y'}^{\text{HBS}}}, \quad \sum_g s_{g,y} = 1, \quad (24)$$

where y' is the HBS wave matched to HICP weight year y . In the baseline, $y' = 2020$ and the expenditure margins are computed from the same household sample as the group baskets. Historical applications select the most recent available preceding HBS wave, or the earliest wave when no preceding observation exists. These targets are expenditure shares, not population shares. For Italy, the income targets use the Istat-based HBS inputs described in Appendix E.

Let $\mathcal{J}_y$ be the set of matched COICOP categories available in the HICP weighting structure in year y . Let $E_{j,g,y'}^{\text{HBS}}$ denote expenditure on item j by group g in the matched HBS wave y' , and let $E_{j,\text{all},y'}^{\text{HBS}}$ denote expenditure on that item by all households. Consistent with the implementation, the HBS correction factor used in the RAS seed is the share of aggregate expenditure on item j

Table F.1: Robustness to rent treatment: inflation gap (in points, 2022)

Country	All rents	Effective rents	Excluding rents
Austria	-0.2	-0.7	0.1
Belgium	2.0	0.5	2.4
Croatia	0.3	0.3	0.4
Cyprus	0.6	0.5	0.7
Estonia	4.5	4.6	4.7
Finland	-1.1	-2.4	-0.9
France	0.1	-0.7	0.2
Germany	0.9	-0.6	1.4
Greece	1.1	0.5	1.3
Ireland	2.8	2.9	3.0
Italy	1.5	1.5	1.6
Latvia	4.6	4.6	4.8
Lithuania	3.3	3.2	3.4
Luxembourg	0.2	-0.8	0.2
Malta	0.2	0.2	0.3
Netherlands	1.8	-0.4	2.7
Portugal	0.2	-0.5	0.3
Slovakia	1.1	0.7	1.2
Slovenia	0.5	0.7	0.5
Spain	0.8	0.3	1.0
Euro area	0.9	0.0	1.2

Notes: All scenarios use national-quintile Eurostat HBS baskets, relative-expenditure weights and chained Laspeyres indices. The first pools HBS 041+042 and maps it to HICP 041; the second uses HBS 041 only; the third excludes HBS 041 and 042 and HICP 041 and renormalises. Gap denotes first-quintile minus fifth-quintile inflation, in percentage points. Euro-area indices aggregate national indices with annual HICP country weights. Italy uses the custom national HBS input; Slovakia uses the 2015 HBS wave.

accounted for by group g :

$$c_{j,g,y'}^{\text{HBS}} = \frac{E_{j,g,y'}^{\text{HBS}}}{E_{j,\text{all},y'}^{\text{HBS}}}. \quad (25)$$

Thus $c_{j,g,y'}^{\text{HBS}}$ records how expenditure on item j is distributed across household groups. To avoid mixing index-weight units with expenditure shares, define the HICP weight normalized over the matched product set as

$$\bar{\omega}_{j,y}^{\text{HICP}} = \frac{\omega_{j,y}^{\text{HICP}}}{\sum_{k \in \mathcal{J}_y} \omega_{k,y}^{\text{HICP}}}, \quad \sum_{j \in \mathcal{J}_y} \bar{\omega}_{j,y}^{\text{HICP}} = 1. \quad (26)$$

The initial product-by-group mass is then

$$Z_{j,g,y} = \max \left\{ s_{g,y} \bar{\omega}_{j,y}^{\text{HICP}} c_{j,g,y'}^{\text{HBS}}, \epsilon \right\}, \quad \epsilon = 10^{-12}. \quad (27)$$

The small positive floor is a numerical safeguard to keep the RAS problem strictly positive after HICP-HBS matching. RAS transforms this seed into a joint expenditure-share matrix $M_{j,g,y}$. Its group margins equal the groups' expenditure shares, while its product margins equal the normalized HICP shares:

$$\sum_{j \in \mathcal{J}_y} M_{j,g,y} = s_{g,y} \quad \text{for all } g, \quad \sum_g M_{j,g,y} = \bar{\omega}_{j,y}^{\text{HICP}} \quad \text{for all } j. \quad (28)$$

Both sets of margins sum to one, so the constraints are mutually compatible.

The RAS algorithm starts from $M_{j,g,y}^{(0)} = Z_{j,g,y}$ and repeats two scaling steps. At iteration t , the group-margin adjustment is

$$\tilde{M}_{j,g,y}^{(t)} = M_{j,g,y}^{(t-1)} \times \frac{s_{g,y}}{\sum_{\ell \in \mathcal{J}_y} M_{\ell,g,y}^{(t-1)}}, \quad (29)$$

and the product-margin adjustment is

$$M_{j,g,y}^{(t)} = \tilde{M}_{j,g,y}^{(t)} \times \frac{\bar{\omega}_{j,y}^{\text{HICP}}}{\sum_h \tilde{M}_{j,h,y}^{(t)}}. \quad (30)$$

We iterate these two steps by weight year until the maximum absolute row or column marginal error is below 10^{-10} , with a maximum of 1,000 iterations.

The calibrated matrix M is a joint expenditure distribution. The group-specific HICP basket is the corresponding conditional product distribution within group g :

$$\omega_{j,g,y} = \frac{M_{j,g,y}}{s_{g,y}} = \frac{M_{j,g,y}}{\sum_{k \in \mathcal{J}_y} M_{k,g,y}}. \quad (31)$$

The equality of the two denominators follows directly from the group-margin constraint. It implies that each group basket sums to one. Moreover, the expenditure-share-weighted average of the group baskets reproduces the normalized HICP product weights:

$$\sum_{j \in \mathcal{J}_y} \omega_{j,g,y} = 1, \quad \sum_g s_{g,y} \omega_{j,g,y} = \bar{\omega}_{j,y}^{\text{HICP}}. \quad (32)$$

The package implements the same system on a percentage scale. Internally, the calibrated mass is $M_{j,g,y}^{\text{pct}} = 100M_{j,g,y}$, so its margins are $100s_{g,y}$ and $100\bar{\omega}_{j,y}^{\text{HICP}}$. The package returns these percentage weights in the R variable `weighted_consumption`. Denoting this variable by $w_{j,g,y}^{\text{code}}$, we have

$$w_{j,g,y}^{\text{code}} = \frac{M_{j,g,y}^{\text{pct}}}{s_{g,y}} = 100\omega_{j,g,y}. \quad (33)$$

This quantity therefore sums to 100 within every household group. Thus division by $s_{g,y}$ in the code is the percentage-scale counterpart of dividing $M_{j,g,y}$ by its group-row sum in the normalized mathematical representation.

This preserves the relative information contained in the HBS expenditure profiles while making the aggregate of the household-group baskets consistent with the published HICP weighting structure, up to residual differences induced by product coverage and COICOP matching.

The all-items reference exercise in Table F.2 compares the two weights for Germany at COICOP digit 3 over the period beginning in 2019. The RAS calibration tracks the published German HICP more closely than the relative-expenditure weighting, whose implicit aggregate weights are equal across groups.

Table F.2: Germany HICP validation by weighting scheme (differences in pp)

Method	Obs.	Mean diff.	Mean abs. diff.	RMSE	Max abs. diff.
RAS	88	-0.059	0.093	0.118	0.259
Relative expenditure	88	-0.070	0.143	0.166	0.449

Notes: Validation statistics compare the recalculated average of German income-quintile inflation rates with the published all-items HICP inflation rate. The sample is monthly, from January 2019 to April 2026. All differences are recalculated minus published inflation and are reported in percentage points.

G Robustness to the Income Definition

The baseline ranks households by total net household income. Table [G.1](#) tests whether this choice drives the measured inflation gap. We keep the household inflation rates, sample, consumption coverage, and weights fixed, and reassign national quintiles using equivalised income.

Equivalisation lowers the euro-area Q1–Q5 inflation gap from 1.4 to 1.1 percentage points in 2022, but does not overturn the aggregate result that lower-income households faced higher inflation. The larger changes in several countries show that household size can affect the composition of national income quintiles and the magnitude of the estimated gap. This pattern is consistent with [Klick and Stockburger \(2024\)](#), who find for the United States that equivalisation substantially changes the demographic composition of income groups but has only a minor effect on their annualised inflation rates. In our setting, the robustness exercise therefore supports the qualitative baseline result while clarifying that total-income quintiles measure household resources, whereas equivalised- income quintiles provide a closer comparison of living standards.

Table G.1: Robustness to income definition: inflation gap (in points, 2022)

Country	Total household income	Equivalised income	Difference
Austria	0.2	0.2	0.1
Belgium	3.2	1.4	-1.8
Croatia	0.3	0.8	0.5
Cyprus	1.0	0.6	-0.4
Estonia	5.0	5.4	0.5
Finland	-0.6	-0.4	0.3
France	0.3	0.1	-0.2
Germany	1.8	1.7	0.0
Greece	1.4	0.7	-0.7
Ireland	4.0	2.1	-1.8
Latvia	5.2	4.0	-1.2
Lithuania	4.6	3.3	-1.4
Luxembourg	-0.1	0.0	0.1
Malta	0.2	0.0	-0.2
Netherlands	3.9	2.6	-1.3
Slovakia	0.1	0.2	0.1
Slovenia	0.6	0.4	-0.2
Spain	1.2	0.8	-0.4
Euro area (18 countries)	1.4	1.1	-0.3

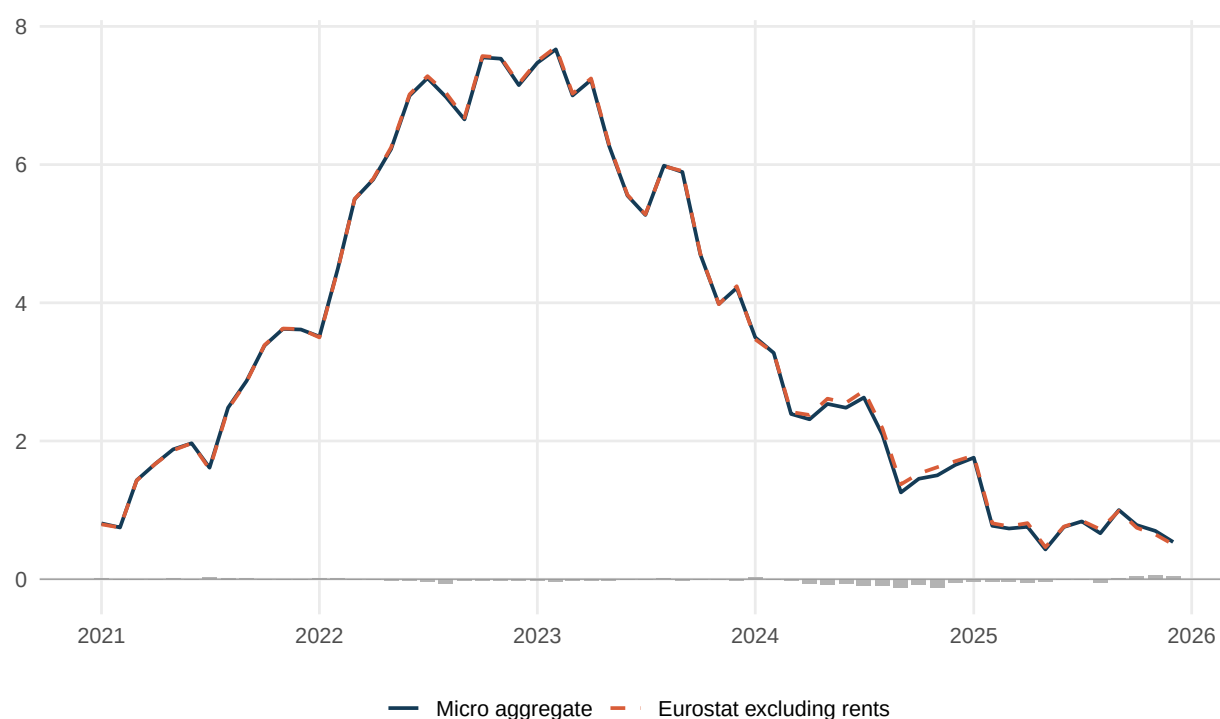
Notes: The table reports first-quintile minus fifth-quintile annual-average household inflation in percentage points. Both columns use the same 2022 household inflation observations, the same rent-excluded consumption universe, RAS weights, survey weights, and country samples. Only the national quintile ranking changes. Total household income is EUR_HH099. Equivalised income divides EUR_HH099 by the OECD equivalence scale HB061. Difference is equivalised income minus total household income. The euro-area row averages national gaps with 2022 HICP country weights over the 18-country sample.

H Validation Tests

Figures H.1 and H.2 compare the microdata indices with Eurostat on matching geographic and non-rent coverage, January 2021–December 2025. The micro aggregate averages December-relative household price links with survey-weighted matched expenditure, then uses annual HICP country weights adjusted for rents before chaining. The household mean instead first averages individual year-on-year inflation with national survey weights, then averages national means with annual all-items HICP country weights normalized over the sample.

H.1 Internal Consistency

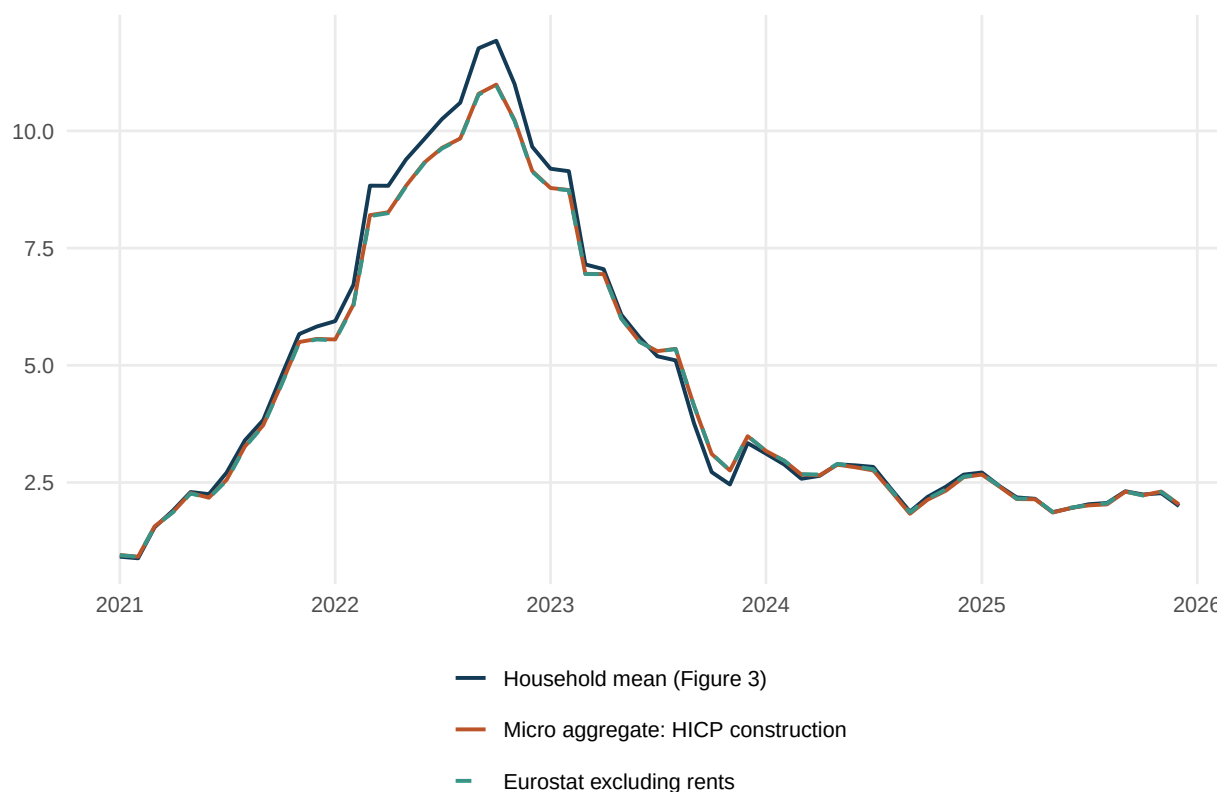
Figure H.1: France: micro aggregate and Eurostat excluding rents (year on year, in %)



Notes: Identical country and time coverage. Grey bars show micro aggregate minus Eurostat, in percentage points. Differences can reflect retained product support and price recoding.

Sources: HBS microdata, Eurostat HICP, authors' calculations.

Figure H.2: Micro–macro reconciliation: 18-country sample excluding rents (year on year, in %)



Notes: All series cover the same 18 countries, excluding Italy and Portugal. The household mean is the Figure 3 series. The two aggregates use identical annual non-rent country expenditure weights and chain price links before calculating inflation.

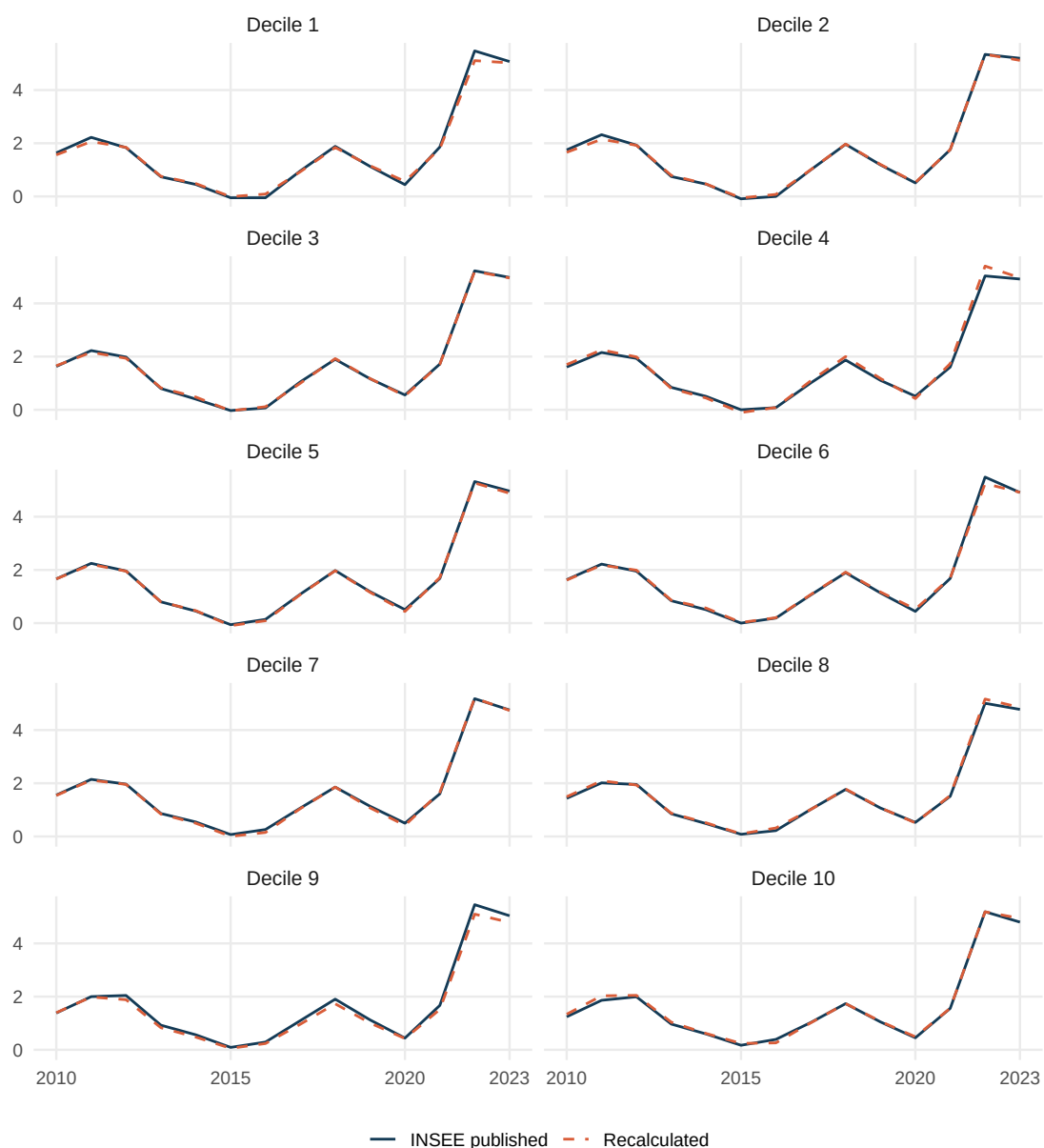
Sources: HBS microdata, Eurostat HICP, authors' calculations.

The mean absolute difference from the Eurostat benchmark is below 0.1 percentage point for the micro aggregate (also below 0.1 at its maximum), compared with 0.2 for the household mean (maximum 1.0). The household mean and expenditure-weighted aggregate measure different objects. The remaining micro-aggregate discrepancy combines product coverage, classification recoding, and price-index construction. Equality on identical product support is partly imposed by RAS calibration and does not independently validate household heterogeneity or policy counterfactuals.

H.2 External Consistency

Figure H.3 retains the historical French comparison with INSEE annual inflation by standard-of-living decile. It uses its original all-items coverage and equivalised-income groups, separately from the non-rent total-household-income baseline.

Figure H.3: External consistency: annual inflation by standard-of-living decile, France (in %)



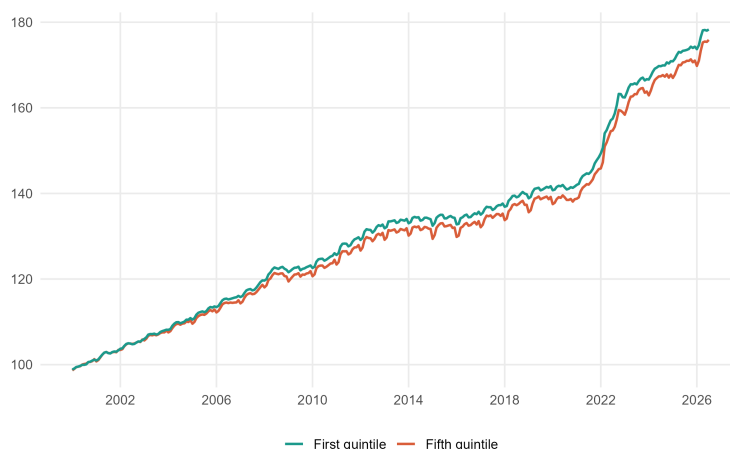
Notes: Annual inflation, 2010–2023. Recalculation uses COICOP digit 4; deciles follow the original INSEE standard-of-living definition. All ten deciles share the same axes.

Sources: INSEE, Eurostat, authors' calculations.

I Long-Run Inflation Inequality

Figure I.1 shows a modest but persistent divergence between the first- and fifth-quintile price indices after 2008. The post-COVID episode produces the largest visible widening in the series since 2000.

Figure I.1: Long-run inflation inequality in the euro area, by income quintile (index, 2000 = 100)



Notes: This figure shows monthly price indices for the first and fifth income quintiles in the euro area, January 2000 to July 2026. Indices are rebased to 2000 = 100.

Sources: Eurostat HICP, HBS, authors' calculations.

This pattern contrasts with the larger long-run gaps documented for the United States by [Jaravel \(2024\)](#). Several methodological differences may account for the contrasting results. First, the euro-area indices reported here aggregate national indices and therefore average across countries whose price dynamics and fiscal responses can offset one another. Second, the euro-area calculation uses harmonized COICOP categories, whereas the U.S. evidence exploits much finer product-level data. Aggregation within broad categories can conceal persistent differences in the prices paid by different income groups, which are captured in the U.S. data.